



DALARNA
UNIVERSITY

Degree Thesis

Master's level

Time Series Analysis for Predicting Air Quality

Author: Puja Khadka

School: School of Information and Engineering

Supervisor: Yves Philippe Rybarczyk and Rasa Zalakeviciute

Examiner: Mia Xiaoyun Zhao

Subject/main field of study: Data Science / Microdata Analysis

Course code: MI4001

Credits: 30

Date of examination: 14 April 2026

At Dalarna University it is possible to publish the student thesis in full text in DiVA. The publishing is Open Access, which means the work will be freely accessible to read and download on the internet. This will significantly increase the dissemination and visibility of the student thesis.

Open Access is becoming the standard route for spreading scientific and academic information on the internet. Dalarna University recommends that both researchers as well as students publish their work Open Access.

I give my consent for full text publishing (freely accessible on the internet, Open Access):

Yes ☒

No ☐

Abstract:

Reliable short- to medium-term air-quality forecasting is important for urban environmental management and public health protection. This study evaluates hourly air-quality forecasting in Quito, Ecuador, using one continuous year of observations from a single monitoring station, from September 1, 2023, to August 31, 2024. Four pollutants were modelled: NO₂, SO₂, CO, and PM_{2.5}. Five forecasting approaches were compared under the same rolling-origin framework: SARIMA, Holt–Winters, VAR, GBM, and a hybrid weather-first SARIMA–GBM model. Each model used a six-month training window, a 72-hour forecast horizon, and lead-wise RMSE evaluation.

The results show that no single model was best for all pollutants. GBM achieved the lowest overall RMSE for NO₂ and CO, with overall RMSE values of 8.35 µg/m³ and 0.231 mg/m³, respectively. The hybrid weather-first model performed best for SO₂, with an overall RMSE of 1.540 µg/m³. PM_{2.5} showed the strongest horizon dependence, with VAR performing best at earlier leads and SARIMA becoming stronger at later leads. Across pollutants, forecast error generally increased with lead time but showed recurring oscillatory behavior, consistent with temporal patterns identified during exploratory analysis.

Overall, the study shows that air-quality forecasting in Quito is pollutant-specific and horizon-sensitive. The findings provide a comparative methodological foundation for future forecasting work, while operational deployment would require real meteorological forecast inputs, drift monitoring, and further validation under practical conditions.

Keywords:

Air quality forecasting, Time series modelling, Rolling-origin evaluation, Gradient Boosting Machine, Vector Autoregression, SARIMA, Holt-Winters, weather-first forecasting, multi-horizon prediction, Quito.

Acknowledgement

I would like to express my sincere gratitude to my supervisor for their valuable guidance, constructive feedback, and continuous support throughout this research. Their expertise and direction were essential in shaping the study and addressing the challenges involved in time-series modelling and air-quality forecasting.

I am grateful to the academic staff of the department for providing a strong theoretical foundation in statistical modelling, machine learning, and environmental data analysis. Their teaching and support greatly contributed to the development of this research.

I would like to acknowledge the organizations responsible for collecting, maintaining, and providing access to the meteorological and air-quality datasets used in this study. Their efforts made this research possible.

I would like to extend my appreciation to Pragya Narsingh Rana for the initial discussions and early-stage support provided during this research.

Finally, I am deeply thankful to my family for their encouragement, patience, and unwavering support throughout my academic journey.

Contents

1.	Introduction	1
1.1.	Air pollution	1
1.2.	Background and Motivation	2
1.3.	Problem statement	3
1.4.	Research objectives	4
1.5.	Research Question	4
1.6.	Research Hypothesis	4
1.7.	Scope of the study	5
2.	Literature review	6
2.1.	Air Pollution and Meteorological Influences	6
2.2.	Univariate statistical models (SARIMA)	6
2.3.	Exponential smoothing models (Holt-Winters)	7
2.4.	Multivariate Time Series Models: Vector Autoregression (VAR)	8
2.5.	Machine-Learning Model Gradient Boosting Machine (GBM)	8
2.6.	Hybrid model (SARIMA-GBM/ Weather First Modelling)	9
2.7.	Comparative synthesis of the literature	11
2.8.	Positioning Relative to Deep Learning Approaches	11
2.9.	Research Gap	11
3.	Data and Study Area	12
3.1.	Study Area	12
3.2.	Air Quality Monitoring Network	13
3.3.	Variables in Dataset	13
3.3.1.	Air Pollutant Data	14
3.3.2.	Meteorological Data	14
3.3.3.	Temporal Structure of the Dataset	14
4.	Methodology	15
4.1.	Overview of methodological workflow	15
4.2.	Data preprocessing and cleaning	15
4.2.1.	Initial Data formatting	16
4.2.2.	One-year window selection	16
4.2.3.	Missing Data Assessment and Imputation	16
4.2.4.	Outlier Handling and Plausibility Constraints	17
4.3.	Exploratory data analysis	17
4.3.1.	Stationarity Testing	18
4.3.2.	Seasonal Decomposition	18
4.3.3.	Meteorology-Pollution Relationships (Pearson and Spearman Correlation)	18

4.3.4.	Temporal Variability Analysis	19
4.4.	Feature Engineering	21
4.5.	Data Modelling	22
4.5.1.	Forecasting Framework overview.....	22
4.5.2.	Rolling Origin Evaluation and Window Configuration	22
4.5.3.	Common Notation and Evaluation Metric	24
4.6.	Model-Specific Implementation.....	25
4.6.1.	Univariate Seasonal Time-Series Forecasting Using SARIMA.....	26
4.6.2.	Univariate Seasonal Smoothing with Holt-Winters	28
4.6.3.	Multivariate Forecasting Using Vector Autoregression (VAR)	30
4.6.4.	Nonlinear Forecasting Using Gradient Boosting Machine (GBM)	32
4.6.5.	Hybrid Weather-First Forecasting using (SARIMA-GBM).....	35
4.7.	Summary of Models and Input Information.....	39
5.	Results and Analysis.....	39
5.1.	Comparative Results Overview.....	39
5.2.	Pollutant Specific Comparative Results.....	42
5.2.1.	Nitrogen Dioxide (NO ₂)	42
5.2.2.	Sulfur Dioxide (SO ₂).....	42
5.2.3.	Carbon Monoxide (CO).....	42
5.2.4.	Fine Particulate Matter (PM _{2.5}).....	42
5.3.	Forecast-Horizon Behaviour Across the 1–72-Hours Windows.....	43
5.4.	Synthesis of Results.....	43
6.	Discussion.....	43
6.1.	Implications of Forecast-Horizon Behaviour	43
6.2.	Why Model Performance Differed Across Pollutants.....	44
6.3.	Practical Implications for Forecasting in Quito	45
6.4.	Study Limitations	45
7.	Conclusion	46
8.	Future Work	47

List of Figures

Figure 1: Air pollution and its effects on human health and the environment.	1
Figure 2: Schematic representation of the variables included in the study dataset.	13
Figure 3: End-to-end methodological workflow of the study.	15
Figure 4: Preprocessing pipeline applied to the selected one-year study dataset.	16
Figure 5: Exploratory data analysis workflow applied to the preprocessed dataset.	17
Figure 6: Pearson correlation heatmap for pollutants and meteorological variables.	18
Figure 7: Spearman correlation heatmap for pollutants and meteorological variables.	19
Figure 8: Representative hourly NO ₂ profile with mean and ± 1 standard deviation.	20
Figure 9: Representative weekly NO ₂ profile with mean and ± 1 standard deviation.	20
Figure 10: Common rolling-origin forecasting and evaluation structure used for all models.	23
Figure 11: SARIMA forecasting workflow within one rolling forecast window.	26
Figure 12: Holt–Winters forecasting workflow within one rolling forecast window.	29
Figure 13: VAR forecasting workflow within one rolling forecast window.	31
Figure 14: GBM forecasting workflow within one rolling forecast window.	34
Figure 15: Hybrid (SARIMA-GBM) forecasting workflow within one rolling forecast window.	36
Figure 16: Lead-wise RMSE for NO ₂ across the five forecasting models.	40
Figure 17: Lead-wise RMSE for SO ₂ across the five forecasting models.	41
Figure 18: Lead-wise RMSE for CO across the five forecasting models.	41
Figure 19: Lead-wise RMSE for PM _{2.5} across the five forecasting models.	41

List of Tables

Table 1: Comparative summary of model families in air-quality forecasting.....	10
Table 2: Summary of hours and weekdays variability for each pollutant.	20
Table 3: Comparison of models on utilization of pollutant and meteorological information. ..	39
Table 4: Best model by pollutant and forecast window based on lead-wise RMSE.	40

1. Introduction

1.1. Air pollution

One of the most significant environmental issues affecting ecosystems, human health, and economic activity is air pollution. It refers to the presence of harmful substances in the atmosphere at concentrations that can impair human health, degrade environmental quality, or interfere with both natural and built systems. According to the World Health Organization (World Health Organization, 2024), air pollution remains a serious global public health threat and causes millions of premature deaths each year. In urban environments, the main anthropogenic sources of pollution include road traffic, fuel combustion, industrial activity, and household emissions, all of which contribute to substantial temporal variability in pollutant concentrations.

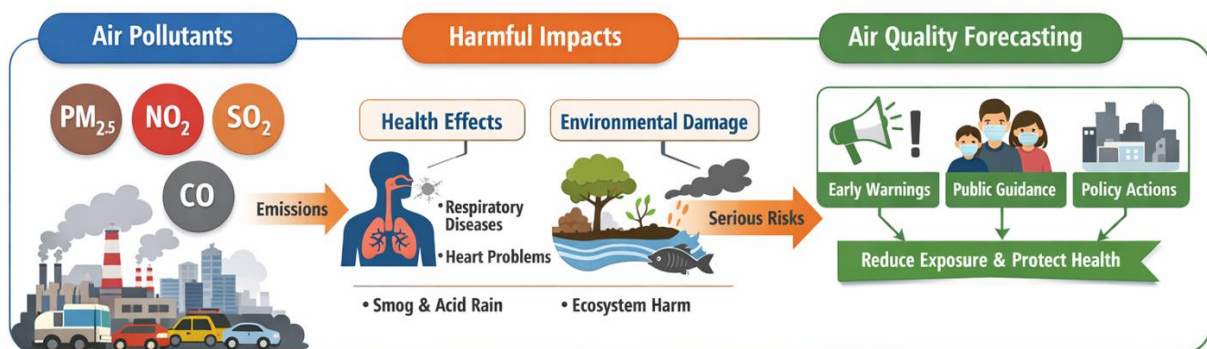


Figure 1: Air pollution and its effects on human health and the environment.

The most commonly monitored pollutants in urban air quality management are ground-level ozone (O₃), carbon monoxide (CO), sulphur dioxide (SO₂), nitrogen dioxide (NO₂), and fine particulate matter (PM_{2.5}). These pollutants are widely studied because of their importance to public health and their frequent inclusion in regulatory monitoring systems. They differ in both their atmospheric behavior and their sources. NO₂ is mainly produced by high-temperature combustion, particularly from traffic and industrial activities. CO is generated by incomplete combustion and is often associated with inefficient fuel use and traffic congestion. SO₂ emissions primarily originate from sulfur-containing fuels and certain industrial processes. PM_{2.5} can arise from both direct emissions and secondary formation in the atmosphere. In contrast to other pollutants, O₃ is a secondary pollutant formed through photochemical reactions driven by solar radiation, volatile organic compounds, and nitrogen oxides.

Exposure to these pollutants is associated with various health risks. Repeated and prolonged exposure has been linked to chronic inflammation, respiratory and cardiovascular disease, and premature mortality. The burden is particularly severe in low- and middle-income countries, where rapid urban growth often coexists with limited environmental regulation and control capacity. As a result, air pollution is not only an environmental problem but also a major public health issue that affects healthcare systems, labor productivity, and overall quality of life.

Air pollution affects more than just human health. It also influences physical infrastructure, agricultural production, climate-related processes, and atmospheric composition. For instance, sulphur-containing pollutants can acidify soils and freshwater systems and accelerate material deterioration, whereas ozone and black carbon contribute to radiative forcing. (Vlachokostas et al., 2010) noted that air pollution can reduce crop yields and lead to economic losses through increased medical costs, reduced worker productivity, and environmental deterioration. These broader impacts highlight the importance of timely and accurate air quality information.

In this study, air pollution is considered both a forecasting issue with immediate local relevance and a broader environmental concern. In a city such as Quito, pollutant concentrations can fluctuate considerably over short periods due to interactions among topography, meteorology, and urban emissions. Therefore, short- to medium-term forecasting can support public communication, short-horizon planning, and early warning systems. Accordingly, this study focuses on hourly predictions of NO₂, SO₂, PM_{2.5}, and CO, which are both locally relevant and suitable for comparative time-series modelling.

The following section builds on this broader understanding of air pollution. It highlights the importance of forecasting, the relevance of the selected pollutants, and the need for a comparative modelling study to identify forecasting approaches suitable for short- to medium-term air-quality prediction in Quito.

1.2. Background and Motivation

Air quality forecasting has grown in importance for environmental management and urban public health. Pollutant levels in densely populated cities can fluctuate quickly in response to weather, combustion sources, and traffic. Therefore, practical measures such as short-term environmental planning, transportation management, and public health alerts can be supported by accurate forecasts. In this way, forecasting serves as both a technical exercise and a decision-support tool to lessen the negative effects of poor air quality on people's health and society (Baklanov & Zhang, 2020).

This requirement is particularly important in Quito, where diverse geography and urban emission sources combine to shape air pollution. The city's high altitude, valley location, limited atmospheric mixing, and frequent temperature inversions can encourage the build-up of pollutants near the surface, and the burden of emissions is increased by traffic growth and urban expansion. Because pollution levels may fluctuate significantly over brief periods and forecasting errors are expected to be affected by both temporal and meteorological variability, Quito is a pertinent case for short- to medium-term forecasting (Rybarczyk & Zalakeviciute, 2021).

However, air quality forecasting remains methodologically difficult. When pollutant behaviour is highly dependent on weather and cross-variable interactions, traditional statistical models are limited in their ability to capture persistence and seasonal structure. Although more adaptable machine-learning techniques can capture nonlinear interactions, they may be challenging to interpret and may require well-designed predictors. By combining time-series modelling with meteorological inputs, hybrid techniques offer a middle ground; nonetheless, their usefulness must be evaluated under the same data requirements and forecast horizons as other model families. This encourages the use of a comparative framework rather than relying solely on one modelling technique (Alleyne et al., 2006).

This need for a controlled evaluation of forecasting techniques in practical short- to medium-term settings is what motivated the current study. Using the same hourly Quito dataset and rolling-origin assessment approach, it contrasts univariate statistical models, exponential smoothing, multivariate linear modelling, nonlinear machine learning, and a hybrid weather-first framework. The emphasis on a 72-hour horizon is both methodological and practical: it is long enough to study how forecast skill varies as uncertainty builds over the forecast window, yet short enough to remain relevant for near-term warning and preparation. For the same reason, the rolling evaluation updates the forecasts every 72 hours to ensure that each cycle offers a full non-overlapping forecast block for uniform model comparison.

The goal of the study is directly related to this background. The study investigates which modelling strategy is best for various pollutants and for different portions of the 1–72 hour short- to medium-term window, rather than whether one forecasting family is always better. In

this sense, the study's motivation is directly related to the subsequent research objectives, questions, and hypotheses.

1.3. Problem statement

Since pollutants are influenced by a variety of interacting factors, such as pollutant persistence, weather conditions, emission patterns, and local topography, forecasting urban air pollution remains challenging. High altitude, valley confinement, limited atmospheric mixing, and frequent temperature inversions all contribute to this problem in a city like Quito, resulting in abrupt daily cycles and transient pollution outbreaks. It is, therefore, methodologically challenging to create forecasting models that are dependable across many pollutants and during a multi-day prediction window.

An additional challenge is that many modelling families have significant constraints and address this issue in different ways. Although they can capture seasonal structure and temporal persistence, univariate statistical models such as Holt-Winters and SARIMA do not explicitly account for meteorological forcing or cross-pollutant dependence. These interactions can be incorporated into multivariate models such as VAR, but they remain constrained by linear assumptions. More intricate nonlinear interactions can be captured by machine-learning models such as GBM, but their effectiveness depends on the availability of appropriate predictor variables and careful feature engineering. By combining time-series forecasting with meteorological data, hybrid techniques offer a promising alternative. However, their comparative effectiveness under consistent forecasting settings remains poorly demonstrated, particularly for cities like Quito.

Therefore, the issue is not only a lack of modelling techniques but rather a lack of clear comparative evidence regarding the best model type under the same data conditions and forecast horizon. A large portion of current research focuses solely on one-step or short-horizon forecasting, uses different datasets and evaluation techniques, or assesses particular approaches separately. Because of this, it is challenging to ascertain whether forecast improvements result from variations in study design or from the modelling family itself. Additionally, this raises a crucial practical question: how does model performance change as the forecast moves from earlier leads to later leads within the 1–72-hour short- to medium-term window?

For Quito in particular, this disparity is crucial. Compared to large cities in North America, Europe, or East Asia, the city is still rarely studied despite its complex terrain, significant weather influences, and unpredictable urban emissions. However, because timely short- to medium-term forecasts can help with public communication, early warning systems, and near-term environmental planning, forecasting in Quito has direct practical value. Knowing whether a model performs well on average is insufficient for these applications; it is also essential to determine whether the model's performance varies by pollutant and by prediction lead time.

Given these factors, the current study examines the need for a controlled comparative evaluation of forecasting methods for Quito's hourly air quality. The study examines univariate statistical models, exponential smoothing, a multivariate linear model, a nonlinear machine-learning model, and a hybrid weather-first framework, all under a shared rolling-origin design, using ground-based hourly pollutant and meteorological data. The study seeks to establish which modelling families are best suited to certain pollutants and how forecast performance varies across the short- to medium-term range by comparing these approaches over the same 72-hour forecast horizon.

1.4. Research objectives

The primary objective of this study is to compare the performance of various time-series forecasting techniques for short- to medium-term air pollution prediction in Quito, Ecuador, using hourly air quality and meteorological data.

Particularly, the current research addresses the following objectives.

1. To assess the performance of univariate forecasting models, especially SARIMA and Holt-Winters, for predicting hourly concentrations of pollutants.
2. To assess whether incorporating meteorological variables and cross-pollutant relationships through a multivariate vector Autoregression (VAR) model improves forecast accuracy compared to univariate models.
3. To assess the forecasting accuracy of a non-linear machine learning model such as Gradient Boosting Machine (GBM), using pollutant history, meteorology data, cross-pollutant relations and temporal features.
4. To create and assess a hybrid weather-first forecasting system that uses meteorological variables as inputs for pollutant prediction after first forecasting them.
5. To assess how forecast accuracy varies with lead time by comparing model performance over forecast horizons ranging from 1 to 72 hours.
6. To identify which modelling technique, given practical data limitations, is best for operational air quality forecasting in Quito.

Together, these goals directly address the methodological perspectives and research gaps, ensuring that the current study advances the field of urban air pollution forecasting from both scientific and practical standpoints.

1.5. Research Question

This study is driven by the following core question:

1. How accurately can short- and medium-term concentrations of key air pollutants in Quito be predicted using univariate statistical models such as SARIMA and exponential smoothing techniques such as Holt-Winters?
2. Does the inclusion of meteorological variables and cross-pollutant dependencies in a multivariate VAR model improve forecast accuracy compared with univariate approaches?
3. Can a non-linear machine learning model, such as GBM, provide better forecasts than statistical models for pollutants influenced by complex temporal and meteorological interactions?
4. How does a hybrid weather-first forecasting framework perform compared with direct statistical, multivariate and machine learning models?
5. How does forecast accuracy change across lead times from 1 to 72 hours, and does the most suitable model vary by pollutant and forecast horizon?

1.6. Research Hypothesis

Both time-dependent and meteorological variability affect air pollutant concentration predictions, and different forecasting models are expected to capture these influences to different degrees. The present study is based on the following hypotheses, which are informed by theoretical predictions and prior empirical research on air-quality forecasting, including studies that highlight the importance of meteorological information for improving prediction accuracy:

Primary Hypothesis (H_1):

For short- to medium-term air-quality forecasting in Quito, models that incorporate meteorological information and multivariate or nonlinear structure are expected to achieve lower forecast error than purely univariate models. In particular, the hybrid forecasting framework is expected to provide a practically valuable approach by combining meteorological forecasting with pollutant prediction under realistic multi-step forecasting conditions.

Null Hypothesis (H_0)

For short- to medium-term air-quality forecasting, models that incorporate meteorological information and multivariate or nonlinear relationships will not achieve better predictive performance than purely univariate models.

1.7. Scope of the study

This study utilises data from a single air-quality monitoring station and collocated meteorological observations to forecast hourly air-pollutant concentrations in Quito, Ecuador, over the short- to medium-term. Only four target pollutants, NO_2 , SO_2 , $\text{PM}_{2.5}$, and CO , are included in the modelling comparison, because they are both health-relevant and compatible with the comparative time-series forecasting framework adopted in this study. CO , NO_2 , and SO_2 are primarily emission-driven pollutants associated with traffic and combustion sources, while $\text{PM}_{2.5}$ is retained despite its greater complexity because it is a key public-health indicator and reflects both primary emissions and secondary formation processes. Including $\text{PM}_{2.5}$ therefore allows the study to examine whether model performance remains consistent for a pollutant with mixed formation pathways.

Ozone (O_3) was excluded from the final modelling scope because it is predominantly a secondary photochemical pollutant formed through nonlinear interactions involving solar radiation, temperature, nitrogen oxides, and volatile organic compounds. Accurately forecasting O_3 would require explicit treatment of photochemical mechanisms and precursor dynamics, which lies beyond the scope of the present time-series and weather-based modelling framework. Including O_3 would therefore reduce methodological comparability across the selected forecasting models.

Although the original dataset contains nearly 20 years of observations, the present study intentionally restricts the analysis to a single continuous one-year period, from September 1, 2023, to August 31, 2024. This choice was methodological, as it provides a controlled and comparable forecasting framework with one complete annual cycle at hourly resolution. It was also motivated by the need to reduce the influence of longer-term structural changes that may affect comparability across years, such as changes in monitoring practices, environmental regulations, emission patterns, and urban conditions. The forecasting scope is restricted to the 1–72-hour horizon explained in Section 4.5.2, which is treated throughout the study as a short- to medium-term prediction window.

The study's methodology is restricted to a comparative analysis of five forecasting techniques: SARIMA, Holt–Winters, VAR, GBM, and a hybrid (SARIMA-GBM) framework. The goal is to compare various modelling families within a single rolling-origin evaluation framework and assess how model adequacy varies by pollutant and forecast lead, rather than creating a fully functional deployment system. As a result, findings should be evaluated based on the location, set of pollutants, modelling framework, and dataset used in this study.

2. Literature review

2.1. Air Pollution and Meteorological Influences

Air pollution is one of the most serious environmental threats to human health worldwide. According to the World Health Organization (World Health Organization, 2024), every year, millions of premature deaths are caused by exposure to air pollution. Fine particulate matter (PM_{2.5}), nitrogen dioxide (NO₂), sulphur dioxide (SO₂), and carbon monoxide (CO) are among the pollutants that pose the greatest threat to public health, as they are all linked to detrimental effects on the heart and lungs. Due to the pronounced temporal variability in pollutant concentrations driven by traffic, combustion processes, and industrial activities, these pollutants are especially prevalent in urban settings. Therefore, accurate forecasting is important for environmental management, public health protection, and scientific understanding.

Meteorological conditions significantly affect air pollution levels by influencing the dispersion, transport, accumulation, and chemical transformation of pollutants. Short-term pollutant behaviour can be strongly affected by factors such as temperature, relative humidity, air pressure, wind direction and speed, precipitation, and solar radiation. (Bai et al., 2018) demonstrate that adding local weather factors significantly enhances urban air-quality prediction, whereas (Alleyne et al., 2006) emphasized that changes in air quality are often difficult to explain without accounting for meteorological variability. While temperature and humidity affect chemical reactions and aerosol generation, wind influences dilution and transport. Taken together, these findings show that pollutant time series cannot be viewed as strictly autoregressive processes, particularly when short- to medium-term forecasting is the aim.

These problems are especially significant in Quito, Ecuador, where complex geography and urban emissions influence air quality dynamics. Quito is situated at a high altitude in a small inter-Andean valley where pollutants can be trapped near the surface due to complex mountain meteorology, frequent temperature inversions, and limited horizontal air movement (Rybarczyk & Zalakeviciute, 2021). Pollutants such as NO₂, SO₂, CO, and PM_{2.5} exhibit significant hourly variations under these conditions. (Cevallos et al., 2017) demonstrated that particulate matter collected in Quito activated inflammatory signaling pathways in human immune cells, supporting the public-health relevance of air pollution exposure in this setting, and the need for accurate forecasting.

Another important methodological consideration is whether the available data are suitable for forecasting. Prior research indicates that high-frequency hourly measurements throughout an entire annual cycle provide a strong basis for short- to medium-term air-quality forecasting because they capture daily, weekly and seasonal variability (Zahedi et al., 2024) and (Zhang & Ding, 2017). This supports the current study's use of a one-year hourly dataset, especially when combined with a rolling-origin architecture that repeatedly re-estimates models using the most recent data. The comparative modelling framework used in this study is therefore motivated by research showing that forecasting air quality in Quito requires attentions to both temporal structure and meteorological forcing.

2.2. Univariate statistical models (SARIMA)

One of the most well-known univariate methods for time-series forecasting in environmental applications is the Seasonal Autoregressive Integrated Moving Average (SARIMA). By adding seasonal components, SARIMA extends the conventional ARIMA framework, making it suitable for data with recurring temporal patterns, such as daily or weekly cycles. This is especially relevant for hourly air-pollution series, where pollutant concentrations frequently

exhibit regular diurnal variation associated with atmospheric mixing, traffic activity, and weather stability. SARIMA is a useful statistical baseline for assessing the predictive skill obtainable from temporal structure alone, as it models a pollutant using its own historical observations.

The application of SARIMA in air-quality forecasting is supported by earlier research. While (Bhatti et al., 2021) reported that SARIMA was able to represent strong seasonal and traffic-related patterns in $PM_{2.5}$ and related pollutants, (Kumar & Goyal, 2011) showed how ARIMA-type models can effectively capture recurring temporal behaviour in urban air-pollution series. These Findings are relevant to Quito, where pollutants such as $PM_{2.5}$, CO, and NO_2 also exhibit distinct daily cycles. Furthermore, (Mahendra et al., 2023) confirmed ARIMA's role as a benchmark in comparative air-quality forecasting.

However, the literature also highlights significant limitations of SARIMA. Because it is essentially a univariate model, it does not account for cross-pollutant interactions or meteorological drivers, both of which significantly affect pollutant concentrations. For instance, (Samal et al., 2019) employed SARIMA as a baseline for pollutant-only forecasting and found that other approaches could perform better in some situations. This suggests that, although SARIMA is effective at capturing seasonality and persistence, it may perform less well when pollutant variability is strongly driven by changing atmospheric conditions rather than by temporal structure alone.

SARIMA is used in this study for two reasons. Firstly, it provides a clear and reliable univariate benchmark for comparison with multivariate, machine-learning, and hybrid models. Second, SARIMA is also used in the hybrid framework, where it forecasts weather-related factors before passing them to the GBM-based pollutant model. As a result, its inclusion is important both as a stand-alone forecasting model and as a component of the hybrid weather-first design used in this study.

2.3. Exponential smoothing models (Holt-Winters)

A popular univariate forecasting technique for time series with level, trend, and seasonality is Holt–Winters exponential smoothing. Its conceptual simplicity, low computational cost, and ability to place greater weight on recent observations make it appealing when modelling environmental variables that change over time. Exponential smoothing forecasts rely on weighted averages of past observations, with weights decreasing exponentially as observations become older (Hyndman & Athanasopoulos, 2021). Because hourly pollutant concentrations frequently exhibit recurrent diurnal patterns linked to traffic activity, atmospheric stability, and boundary-layer dynamics, Holt–Winters is relevant for air-quality forecasting. When seasonal fluctuations remain approximately constant in amplitude over time, the additive Holt–Winters specification is often appropriate.

Holt–Winters is a useful baseline for air-quality forecasting, particularly when the goal is to capture short-term seasonal structure in a transparent and computationally efficient manner. (Jayanth et al., 2025) showed that Holt-Winters smoothing can successfully extract level, trend, and seasonal structure prior to more complex modelling stages, while (Malik et al., 2020) showed that Holt–Winters can provide reliable short-term forecasts for air-quality indicators. These findings suggest that Holt–Winters is valuable as a seasonal baseline against which more complex models can be evaluated, rather than necessarily as the most advanced forecasting technique.

However, Holt–Winters also has significant drawbacks. As a univariate smoothing method, it does not directly account for cross-pollutant relationships or meteorological variables, which can be important in air-quality forecasting. In addition, its robustness may be reduced in complex environmental series because of sensitivity to missing or irregular data and to the

choice of smoothing parameters (Rumbe et al., 2024). For this reason, Holt–Winters is used primarily as a baseline model in the present study rather than as the expected best-performing approach.

2.4. Multivariate Time Series Models: Vector Autoregression (VAR)

A multivariate time-series framework called Vector Autoregression (VAR) models each variable as a function of its own lagged values and the lagged values of all other variables in the system. This is helpful for air-quality forecasting because pollutant concentrations are sometimes influenced by other pollutants and climatic circumstances in addition to their own historical data. Thus, VAR may capture joint temporal dynamics and cross-variable dependence within a single statistical framework, in contrast to univariate models such as SARIMA and Holt-Winters. Because of this, it provides a good middle ground between straightforward univariate techniques and more adaptable nonlinear machine-learning models.

The use of VAR as a significant multivariate baseline is supported by data from air quality studies. (Abdul Rahim & Masseran, 2021) demonstrated that the model could capture interdependence among pollutant series while generating short-term projections over several days by applying VAR to a variety of pollutants in Malaysia. Their application of Granger causality analysis, lag selection, and stationarity testing demonstrates how VAR can be methodically applied to urban air pollution data. Even when later integrated with machine-learning techniques, (Yi & Lin, 2025) discovered that VAR offered a robust statistical description of temporal pollutant behaviour. These studies collectively imply that VAR is useful when shared temporal and meteorological structures, rather than isolated single-series dynamics, shape pollutant evolution.

The interpretability of VAR is one of its main advantages. Researchers may investigate how changes in one variable may affect others over time since the model clearly incorporates lagged interactions between pollutants and meteorological factors. Because pollutants frequently share emission sources and react to atmospheric conditions together, this is very helpful in air-quality forecasting. According to (Ali et al., 2025) VAR frameworks facilitate the interpretation of dynamic linkages in environmental systems through the use of methods like Granger causality and impulse-response analysis. This makes VAR a crucial multivariate benchmark for the current investigation, to assess whether explicitly modelling weather and cross-pollutant dependencies enhances forecasting performance compared with univariate methods.

VAR has certain drawbacks despite these benefits. The assumption of linear connections, which may not fully reflect the nonlinear behaviour frequently found in pollution generation, dispersion, and climatic interactions, is its primary flaw (Yi & Lin, 2025). Additionally, because model dimensionality increases rapidly, VAR models can be challenging to estimate when numerous variables and lag terms are added, particularly with hourly data. In the context of the current study, where pollutant dynamics may involve both multivariate dependency and nonlinear effects, these constraints are significant. Because of this, VAR is included as a statistically supported multivariate reference model for comparing machine-learning and hybrid approaches, rather than as a final solution.

2.5. Machine-Learning Model Gradient Boosting Machine (GBM)

Machine-learning approaches have become increasingly important in air-quality forecasting because pollutant concentrations are often shaped by nonlinear interactions among meteorological conditions, pollutant history, and temporal patterns. Such relationships can be difficult to capture using simple univariate or linear multivariate statistical models. Gradient

Boosting Machines (GBM) have attracted particular interest because they reduce forecast error by progressively adding weak learners, typically decision trees. This makes GBM well-suited to structured environmental data, where nonlinear effects may be important but the dataset remains tabular rather than textual or image-based (Balasubramanian, 2022) and (Fetahi et al., 2025).

According to the literature, GBM is a strong option for air-quality forecasting. (Rybarczyk & Zalakeviciute, 2021) demonstrated that GBM could detect important relationships between air quality and meteorological conditions in a topographically complex metropolitan area by estimating business-as-usual pollution levels in Quito. GBM is also highlighted by (Chau et al., 2022) as an important component of weather-related modelling approaches that use meteorological predictors to estimate pollutant concentrations under changing environmental conditions. More generally, (Fetahi et al., 2025) indicate that gradient-boosting models perform well in short-horizon forecasting when combined with temporal feature engineering, including cyclical encodings, rolling statistics, and lagged values.

One of GBM's main advantages is its ability to capture nonlinear interactions without requiring a strict parametric structure. This is particularly valuable when predicting less frequent but operationally important high-pollution episodes. These characteristics make GBM the main nonlinear machine-learning benchmark in this study.

However, GBM also has limitations. Without additional diagnostic tools, it may be less interpretable than classical statistical models. Its performance also depends on careful hyperparameter tuning and appropriate feature engineering. A further difficulty arises in multi-step forecasting: GBM performance may decline over longer horizons if future predictors, especially meteorological inputs, are not directly observed. This limitation directly motivates the inclusion of the hybrid (SARIMA-GBM) framework in the present study.

2.6. Hybrid model (SARIMA-GBM/ Weather First Modelling)

Hybrid forecasting approaches have become important in air-quality prediction because they combine the strengths of statistical time-series models and machine-learning methods. Their motivation comes from the fact that pollutant concentrations often exhibit both nonlinear dependence on meteorological conditions and structured temporal behaviour such as seasonality and autocorrelation. A single modelling technique may struggle to capture both aspects adequately. Hybrid models therefore divide these tasks by allowing one component to model temporal structure and another to model nonlinear pollutant–weather relationships.

In the broader literature, Weather Normalised Modelling (WNM) often refers to approaches that account for meteorological variability in order to separate weather effects from emission-related changes in pollutant concentrations. (Alleyne et al., 2006) describe weather-related approaches of this kind as a way of accounting for meteorological effects when analysing or forecasting pollution concentrations. In some implementations, time-series techniques such as ARIMA or SARIMA are first used to model or forecast meteorological variables, and the resulting meteorological inputs are then passed to a machine-learning model, often a GBM, for pollutant prediction. This structure is appropriate because the statistical component can capture regular temporal patterns in meteorology, while the machine-learning component can model nonlinear relationships between meteorological conditions, time-related factors, and pollutant levels.

In an operational forecasting setting where future meteorological observations are not immediately available at prediction time, this type of hybrid design can be particularly useful. (Chau et al., 2022) demonstrate that weather-related modelling approaches can be useful in Quito, where complex topography and meteorological variability strongly influence pollutant behaviour. More broadly, hybrid (SARIMA-GBM) frameworks address a major limitation of

direct machine-learning approaches: although GBM can model nonlinear pollutant responses, it cannot generate future meteorological inputs on its own. By forecasting meteorology first and then using those forecasts to predict pollutant concentrations, hybrid models provide a more operationally realistic multi-step forecasting framework for short- to medium-term air-quality prediction.

However, hybrid models also introduce new challenges. Their performance depends on controlling the propagation of errors from meteorological forecasts into pollutant forecasts, appropriately handling lagged structure, and maintaining temporal alignment between modelling stages. In other words, any error introduced in the first-stage meteorological forecast may affect the second-stage pollutant forecast. For this reason, hybrid models cannot automatically be assumed to outperform all other approaches.

In this study, the hybrid model is not used as a pure deweathering or normalisation method. Instead, it is implemented as a weather-first forecasting pipeline, in which meteorological variables are forecast first using SARIMA and then used as inputs to a GBM model to predict pollutant concentrations. This distinction is important because it clarifies that the study draws on the broader WNM literature while applying the concept in a more specific forecasting context. The role of the hybrid model in this study is therefore both methodological and practical: it is used to examine whether explicitly forecasting meteorology before pollutant prediction improves short- to medium-term forecasting performance under realistic operational constraints.

Table 1: Comparative summary of model families in air-quality forecasting.

Model family	SARIMA	Holt-Winters	VAR	GBM	Hybrid
Main Strength	Captures autocorrelation seasonality in single-pollutant series	Simple, fast, and effective for recurring seasonal structure	Models joint dynamics among pollutants and meteorological variables in a single multivariate framework	Captures nonlinear interactions among pollutant history, meteorology, and temporal features	Provides an operationally realistic framework by forecasting meteorology before pollutant prediction
Main limitation	Does not directly model meteorological effects or cross-pollutant interactions	Limited flexibility and no direct use of meteorological covariates	Assumes linear relationships and can become unstable in high-dimensional settings	Requires careful feature engineering and may degrade in recursive multi-step forecasting when future predictors are unavailable	Sensitive to upstream meteorological forecast error and stage-to-stage alignment
Typical usefulness	Useful as a strong univariate	Useful as a transparent seasonal	Useful when pollutant behaviour is	Strong candidate for short- to	Valuable when future meteorologic

in 1-72h Forecasting	baseline, especially when temporal persistence and diurnal structure dominate	baseline for short-horizon forecasting	shaped by shared temporal and meteorological structure	medium-term forecasting with structured tabular predictors	all observations are unavailable at prediction time
Relevance to the study	Included as a univariate statistical benchmark and as the meteorological forecasting component in the hybrid framework	Included as a simple exponential-smoothing benchmark	Included as the main multivariate linear benchmark	Included as the main nonlinear machine-learning model	Included to test whether a two-stage weather-to-pollutant framework improves practical forecasting performance

2.7. Comparative synthesis of the literature

Collectively, the literature indicates that no single forecasting model consistently performs best across all air-quality forecasting tasks. Univariate models such as SARIMA and Holt–Winters remain important because they capture persistence, seasonality, and recurrent diurnal structure, and they often provide competitive baselines at short lead times when pollutant behaviour is relatively regular. However, their effectiveness is usually limited when pollutant concentrations are strongly influenced by weather or by interactions with other variables.

For forecast horizons within approximately 1–72 hours, the literature suggests that simple univariate models can remain competitive when persistence and seasonality dominate, whereas machine-learning and hybrid approaches tend to become more advantageous when meteorological forcing, nonlinear interactions, and multi-step forecasting constraints play a larger role. The comparative methodology adopted in the present study is therefore justified by prior research showing that model suitability depends on forecast horizon, pollutant characteristics, and the importance of meteorological forcing.

2.8. Positioning Relative to Deep Learning Approaches

Recent air-quality forecasting research has also employed deep-learning architectures such as LSTM, GRU, Temporal Convolutional Networks (TCN), and Transformer-based models to capture nonlinear and long-range temporal dependencies. These methods can achieve strong predictive performance, especially when richer spatial inputs, multi-station data, or longer training histories are available.

However, these models were excluded from the present study to maintain a controlled, methodologically consistent comparison among SARIMA, Holt–Winters, VAR, GBM, and the hybrid framework. The analysis was deliberately limited to a one-year hourly window from a single station, and the aim was not to exhaustively evaluate every possible forecasting architecture. Instead, the study focuses on comparing statistical, multivariate, machine-learning, and hybrid approaches under the same rolling-origin, leakage-safe evaluation design.

2.9. Research Gap

The reviewed literature shows major advances in air-quality forecasting using machine-learning models, hybrid frameworks, multivariate time-series approaches, univariate statistical

models, and exponential smoothing techniques. Earlier studies indicate that each modelling family can perform well in particular situations, especially for short-term forecasting and in settings where meteorology strongly affects pollutant behaviour. However, several important gaps remain.

First, only a small number of studies offer a systematic evaluation of multiple forecasting families within a single experimental framework. It is difficult to assess the relative strengths and weaknesses of statistical, machine-learning, and hybrid models under the same data conditions, prediction horizon, and evaluation methodology because much of the current literature examines these models separately. There are still a few studies that compare nonlinear machine-learning models such as GBM, multivariate approaches such as VAR, univariate models such as SARIMA and Holt–Winters, and hybrid approaches within the same dataset.

Second, instead of using high-resolution hourly data from ground-based monitoring stations, many air-quality forecasting studies rely on aggregated indicators, daily averages, or satellite-derived outputs. Although such data are useful for broader regional analysis, they may not capture the fine-scale temporal variability required for operational forecasting and local public-health applications.

Third, while a great deal of research has been conducted on short-term forecasting, fewer studies have examined how predictive performance changes consistently across a range of lead times. Without evaluating how forecast error changes over longer horizons, many studies present results for only one forecast step, such as one hour or twenty-four hours ahead. As a result, comparative evidence on the behaviour of different model families across lead times from immediate forecasts to multi-day predictions remains limited.

Fourth, although the term Weather Normalised Modelling (WNM) is used in different ways in the literature, hybrid approaches inspired by WNM have shown considerable promise. Broadly speaking, WNM often refers to methods that account for meteorological variability to separate weather-related effects from emission-driven changes in pollutant concentrations. In the present study, however, the hybrid framework uses the idea in a more specific forecasting sense: meteorological variables are first forecasted using SARIMA, and a GBM model then uses those forecasted variables to predict pollutants. Therefore, the hybrid model in this study is treated as a weather-first forecasting pipeline rather than as a pure deweathering or normalisation technique. Comparisons of this kind of hybrid pipeline with univariate, multivariate, and direct machine-learning models remain limited, particularly across forecast horizons and pollutant types.

Lastly, the literature also contains a geographical gap. Most air-quality forecasting research has been conducted in North America, Europe, and East Asia, whereas high-altitude and topographically constrained environments such as Quito remain underrepresented.

Using hourly ground-based air-quality and meteorological data from Quito, the present study compares SARIMA, Holt–Winters, VAR, GBM, and a hybrid (SARIMA-GBM) forecasting framework in order to address these gaps. By using a shared rolling-origin design and assessing forecasts over a 72-hour horizon, the study aims to clarify how different modelling families perform under consistent conditions and which methods are best suited to short- to medium-term air-quality forecasting in a complex urban environment.

3. Data and Study Area

3.1. Study Area

The capital of Ecuador and one of the world's highest-altitude urban centers, Quito, is the subject of this research. Quito resides in a small inter-Andean valley in the northern Andes, at a typical elevation of around 2,850 meters above sea level. The city's hilly surroundings

significantly affect the local climate and air circulation patterns. These geographical factors significantly influence the distribution, accumulation, and temporal variability of air pollutants.

Because of its complex topography, Quito frequently experiences temperature inversions, especially at night and in the early mornings, which limit vertical air mixing and encourage the buildup of pollutants near the ground. Taken together with comparatively poor horizontal air mixing during certain periods of the year, these factors may result in noticeable diurnal oscillations in pollution. Vehicle traffic is the main source of urban pollution in Quito, with household combustion sources and industrial activity also contributing. Because of these features, Quito is an exceptionally pertinent environment for assessing air-quality forecasting techniques, particularly those intended for short- to medium-term operational use.

3.2. Air Quality Monitoring Network

Quito's municipal air-quality monitoring network consists of numerous fixed monitoring stations distributed across regions with varying elevations, population densities, and emission conditions. Although the larger network offers a broader spatial perspective, the current study uses data from a single station in order to focus on temporal forecasting under controlled conditions.

One station was chosen for methodological reasons. By limiting the analysis to a single monitoring location, the study isolates the temporal and meteorological aspects of forecasting performance, thereby avoiding the introduction of regional variation in emissions, land use, and station-specific characteristics. This is appropriate for the current study, which aims to compare forecasting models rather than develop a framework for spatial prediction.

3.3. Variables in Dataset

The dataset used in this study consists of hourly observations indexed by a common timestamp and includes three primary categories of variables: pollutant, meteorological, and temporal data. The variables included in the dataset are illustrated in Figure 2.

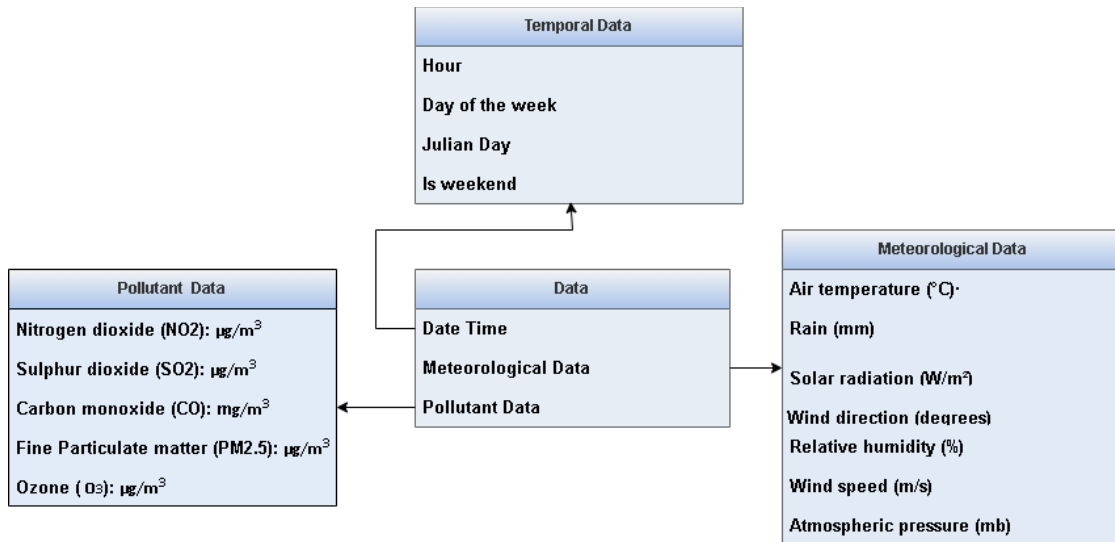


Figure 2: Schematic representation of the variables included in the study dataset.

3.3.1. Air Pollutant Data

The dataset consists of hourly observations of air pollution concentrations collected at a single Quito air quality monitoring station. The following pollutants were examined in this study:

- Carbon monoxide (CO) mg/m^3
- Nitrogen dioxide (NO_2) $\mu\text{g/m}^3$
- Sulfur Dioxide (SO_2) $\mu\text{g/m}^3$
- Fine Particulate matter ($\text{PM}_{2.5}$) $\mu\text{g/m}^3$
- Ozone (O_3) $\mu\text{g/m}^3$

For modelling, the analysis focused on CO, NO_2 , SO_2 , and $\text{PM}_{2.5}$. O_3 was retained for descriptive inspection during exploratory analysis but was not included in the final forecasting comparison, as explained in the Section 0.

3.3.2. Meteorological Data

The air quality data were correlated with weather variables captured over the same time frame. The dataset includes the following weather variables:

- Air temperature ($^{\circ}\text{C}$)
- Relative humidity (%)
- Atmospheric pressure (mb)
- Solar radiation (W/m^2)
- Wind speed (m/s)
- Wind direction (degrees)
- Precipitation (mm)

These variables are known to affect the transformation, accumulation, and dispersal of pollutants. Their inclusion is crucial for the multivariate, machine-learning, and hybrid modelling techniques examined in this study.

3.3.3. Temporal Structure of the Dataset

The monitoring dataset was recorded at hourly resolution and indexed by the Date_Time variable. This timestamp structure was important because it allowed each observation to be linked not only to pollutant and meteorological measurements, but also to recurring temporal information such as hour-of-day, day-of-week, day-of-year, and weekend status.

These temporal attributes were not treated as separate measured environmental variables, but were derived from the timestamp during preprocessing and feature engineering. Hour-of-day information was used to represent diurnal variation, day-of-week information was used to represent weekly patterns, and day-of-year information was used to capture broader within-year seasonal change. A weekend indicator was also created to distinguish regular weekdays from weekends.

The selected one-year period provides a continuous hourly structure that supports the later exploratory analysis of diurnal and weekly variability, as well as the construction of cyclical temporal encodings used in the GBM, VAR, and hybrid modelling stages. The methodological rationale for restricting the analysis to this one-year window is provided in Section 1.7.

4. Methodology

4.1. Overview of methodological workflow

This section covers the methodological approach used to create the forecasting models, prepare the dataset, and assess their performance using a standard experimental design. A structured workflow comprising data loading and selection, pre-processing, exploratory data analysis, feature engineering, forecasting model development, and comparative evaluation was used to analyse a one-year hourly dataset selected from the larger Belisario monitoring archive.

The study compares five forecasting techniques. All models were assessed using the same rolling-origin design and evaluation metrics to ensure comparability. Figure 3 summarises the study's complete procedure.

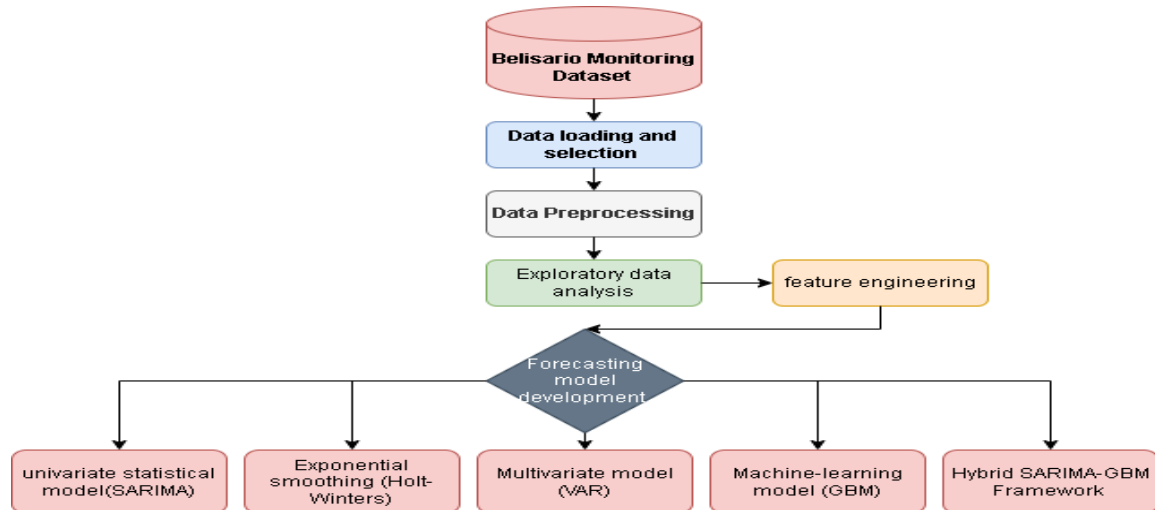


Figure 3: End-to-end methodological workflow of the study.

Following dataset selection, the dataset was cleaned and transformed into an hourly time-series format suitable for the modelling. Next, seasonal trends, temporal variability, and pollutant–meteorology correlations were identified by exploratory research. Then, considering the findings, feature engineering was performed, and the forecasting models were implemented. Model evaluation and comparison across pollutants and Forecast lead durations comprised the last phase.

4.2. Data preprocessing and cleaning

The raw hourly dataset was converted into a model-ready time-series dataset suitable for statistical, multivariate, machine-learning, and hybrid forecasting after a one-year study window was selected from the original dataset. Data loading and initial formatting, restriction to the selected one-year research window, missing-data evaluation and imputation, and outlier handling with physical-plausibility constraints comprised the sequential workflow of the preprocessing stage. The purpose of this sequence was to ensure that the dataset used for

subsequent analysis was suitable for model development, numerically consistent, physically plausible, and temporally coherent.

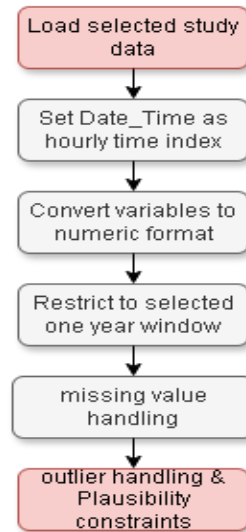


Figure 4: Preprocessing pipeline applied to the selected one-year study dataset.

There were two primary functions of the pre-processing phase. By setting the datetime index, translating variables to numeric format, and limiting the analysis to the selected annual study period, it first created a clear and consistent hourly time-series structure. Second, it used variable-appropriate cleaning techniques to address frequent data-quality issues in environmental monitoring series, such as handling missing values and outliers. After that, the dataset was utilised for exploratory analysis, feature engineering, and the deployment of the forecasting model.

4.2.1. Initial Data formatting

The raw monitoring data were first loaded from Excel format into a pandas DataFrame for inspection and processing. The Date_Time column was converted to a datetime format and set as the hourly time index to support chronological sorting, temporal slicing, interpolation, lag construction, and rolling-origin analysis. After the time index was established, all remaining variables were converted to numeric format, and any corrupted or non-numeric entries were coerced to missing values (NaN). This step ensured that the dataset had a consistent numerical and temporal structure before further preprocessing was applied.

4.2.2. One-year window selection

Following initial formatting, the dataset was restricted to the selected one-year analysis window, from September 1, 2023, to August 31, 2024, by filtering the time index. All subsequent preprocessing steps, including missing-value handling, outlier treatment, and feature engineering, were applied exclusively within this restricted dataset. The rationale for selecting this time window is discussed in the Section 0.

4.2.3. Missing Data Assessment and Imputation

Missingness was assessed for each variable by calculating the number and proportion of missing observations. This stage was used to determine the extent of incomplete data and to guide the choice of imputation method. Because the variables differed in their temporal behaviour, missing values were not handled using a single generic procedure.

Imputation was carried out using variable-appropriate methods. Time-based interpolation, which is appropriate for series that typically evolve smoothly at hourly resolution, was used to

fill missing values in the weather variables and CO. To maintain recurrent diurnal structure, missing values for the remaining pollution variables were handled using STL-based interpolation with a daily seasonal period. Following these procedures, any remaining gaps were filled using KNN imputation with $k = 5$. This sequence was designed to minimise the possibility that missing-data patterns would distort subsequent exploratory research and model development while maintaining temporal continuity.

When necessary, some models used training-window-only missing-value handling during rolling forecasting in addition to this dataset-level pre-processing. In such cases, the entire imputation pipeline was not repeated. Instead, this step served as a methodological safeguard to ensure that any residual inaccuracies or incompleteness in a given training slice were handled using only data available prior to the prediction origin. This distinction is crucial for leakage-safe time-series forecasting because model-time data preparation must not use observations from the prediction horizon.

4.2.4. Outlier Handling and Plausibility Constraints

After missing-value imputation, extreme and physically implausible observations were addressed in the dataset. This step was necessary because environmental monitoring data may contain negative values, isolated spikes, or sensor-related artefacts that are inconsistent with realistic pollutant behaviour and that could disproportionately affect model fitting.

Accordingly, outlier handling was applied in a limited and controlled manner. Since negative pollutant concentrations have no physical meaning, they were trimmed to zero. Winsorization at the first and 99th percentiles were then used to reduce extreme values, and cautious upper caps were applied to exclude unusual peaks that might represent measurement error rather than actual atmospheric conditions. The objective of these operations was to preserve the underlying temporal variability of the series while reducing the influence of implausible extremes.

4.3. Exploratory data analysis

After pre-processing, exploratory data analysis was conducted to describe the dataset's temporal structure and identify connections relevant to model development. Stationarity testing, seasonal decomposition, pollutant–meteorology relation, and temporal variable analysis were the main components of the EDA stage. To determine whether the cleaned series were appropriate for time-series modelling and to guide subsequent feature engineering and forecasting decisions, the studies were combined. Figure 5 summarises the entire EDA workflow.

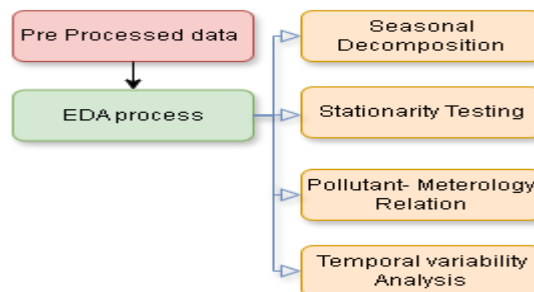


Figure 5: Exploratory data analysis workflow applied to the preprocessed dataset.

4.3.1. Stationarity Testing

Stationarity was assessed using the Augmented Dickey-Fuller (ADF) test for both pollutant and meteorological variables. This step was included to examine whether the cleaned time series were sufficiently stable over the selected one-year window for time-series modelling, particularly for models such as SARIMA and VAR. For all tested variables, the ADF p-values were below 0.05, indicating rejection of the ADF unit-root null hypothesis within the selected period. This suggests that the series was stationary enough for the intended forecasting analysis. However, this result does not imply the absence of diurnal or seasonal structure; rather, it indicates that the series did not show evidence of an uncontrolled stochastic trend over the analyzed window. Detailed test results are provided in Appendix.

4.3.2. Seasonal Decomposition

An additive formulation with a daily seasonal period of 24 hours was used for seasonal decomposition. The objective was to confirm the existence of a recurring daily structure and to divide each hourly series into trend, seasonal, and residual components. Because it helped distinguish regular temporal variation from irregular fluctuations and provided methodological support for the use of seasonal models such as SARIMA and Holt-Winters, this analysis was particularly crucial for pollutants exhibiting significant diurnal cycles.

4.3.3. Meteorology-Pollution Relationships (Pearson and Spearman Correlation)

Both Pearson and Spearman correlation analyses were used to look at the relationships between pollutants and meteorological factors. While monotonic correlations that might not be strictly linear were evaluated using Spearman's correlation, linear relationships were assessed using Pearson's correlation. The purpose of this stage was to assess the likelihood that meteorological predictors will provide valuable data to multivariate, machine-learning, and hybrid forecasting models.

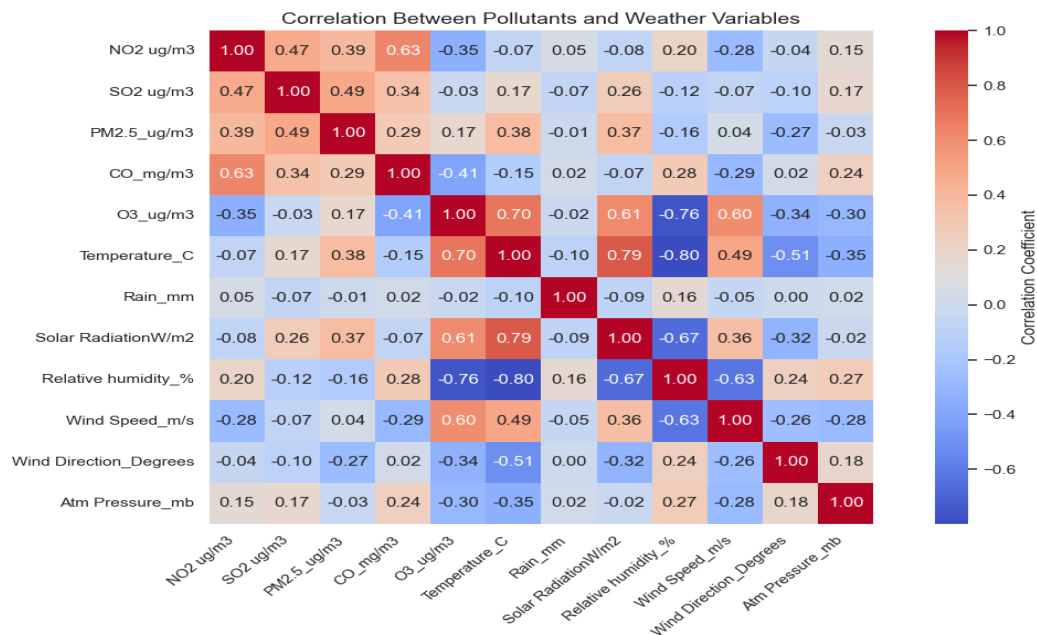


Figure 6: Pearson correlation heatmap for pollutants and meteorological variables.

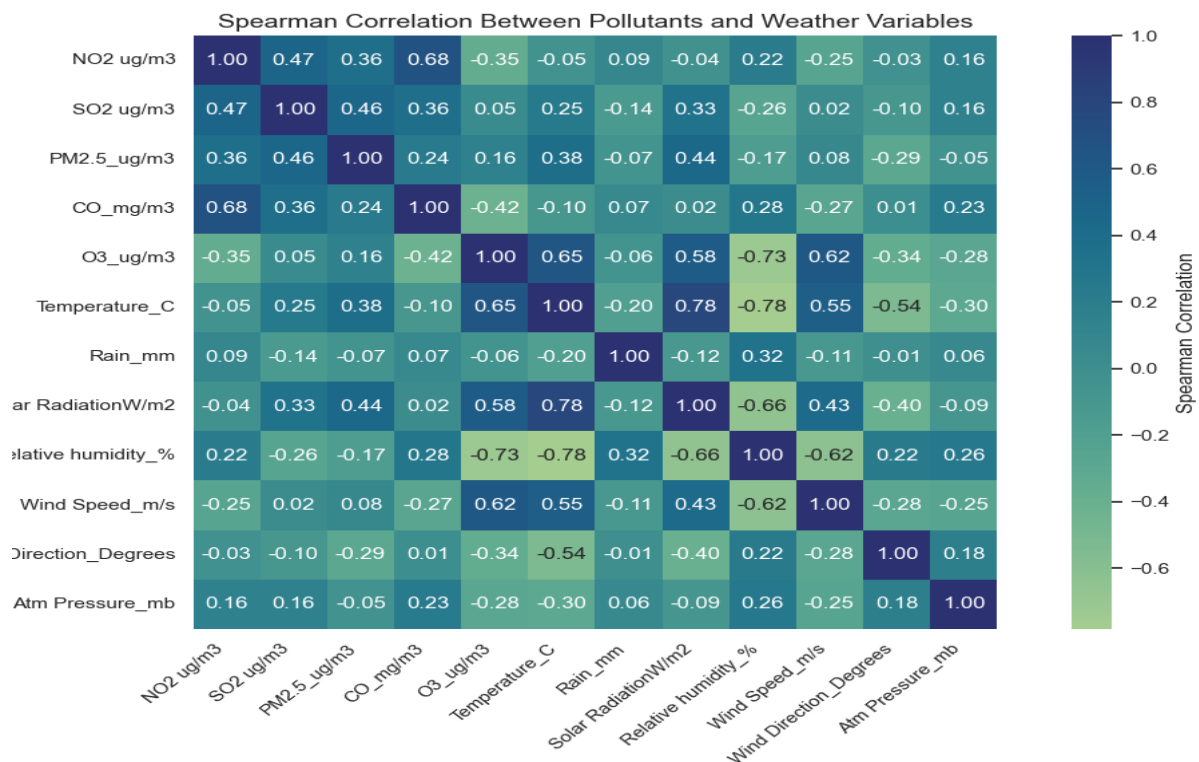


Figure 7: Spearman correlation heatmap for pollutants and meteorological variables.

The meteorological variables showed the strongest internal relationships, indicating physically coherent weather behaviour. For example, there was a strong positive correlation between temperature and solar radiation (Pearson ≈ 0.79 , Spearman ≈ 0.78), and a strong negative correlation between relative humidity and both temperature (Pearson ≈ -0.80 , Spearman ≈ -0.78) and solar radiation (Pearson ≈ -0.67 , Spearman ≈ -0.66). In addition, wind speed was negatively correlated with relative humidity (Pearson = -0.63 , Spearman ≈ -0.62), which is consistent with normal midday drying and ventilation behaviour.

Among pollutant relationships, NO₂ and CO showed a substantial positive correlation (Pearson = 0.63 , Spearman ≈ 0.68), suggesting common traffic-related emission sources and similar accumulation behaviour under stagnant atmospheric conditions. Within this dataset, PM_{2.5} showed a moderate positive correlation with solar radiation (Pearson = 0.37 , Spearman ≈ 0.44) and temperature (Pearson ≈ 0.38 , Spearman ≈ 0.38), indicating that PM_{2.5} variability may reflect both emission patterns and meteorologically mediated processes. NO₂ showed a negative correlation with wind speed (Pearson = -0.28 , Spearman ≈ -0.25), which is physically consistent with dispersion effects in which stronger winds reduce near-surface accumulation.

4.3.4. Temporal Variability Analysis

To characterise recurrent temporal structure in the pollutant series, hour-of-day and day-of-week summary statistics were analysed during the exploratory stage. For each pollutant, mean concentration profiles and corresponding standard deviations were computed across hours of the day and days of the week. The mean profiles capture systematic diurnal and weekly

patterns, while the standard deviations indicate periods of relatively low and high variability, providing insight into how forecasting difficulty may change across different temporal regimes.

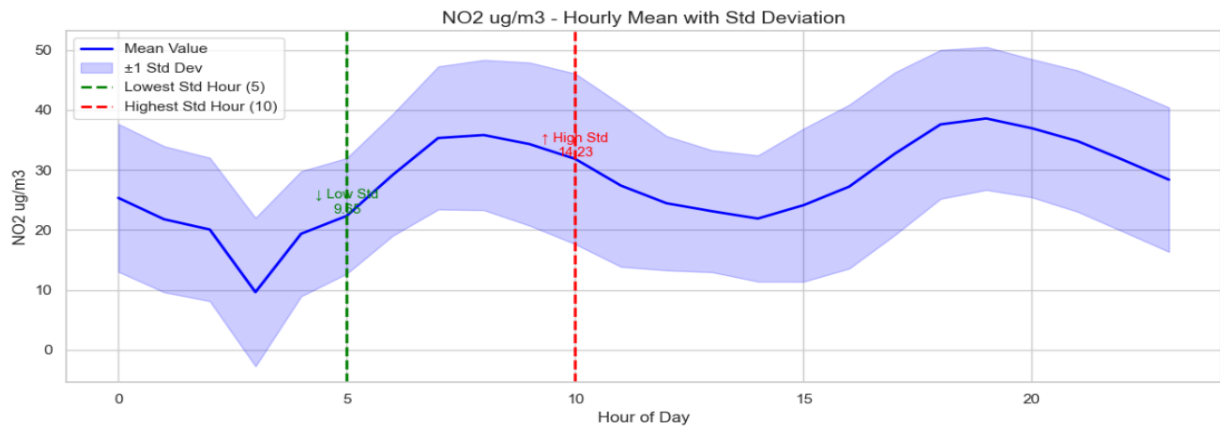


Figure 8: Representative hourly NO_2 profile with mean and ± 1 standard deviation.

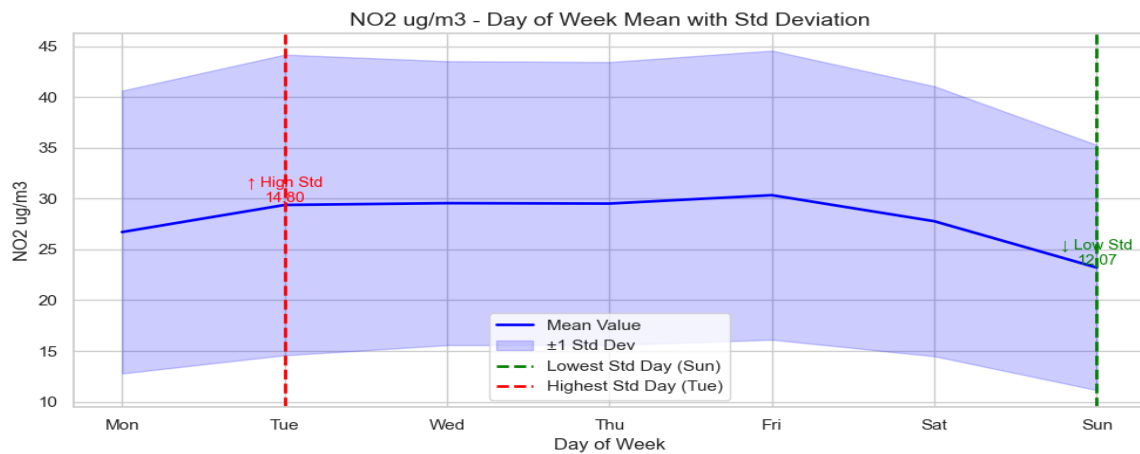


Figure 9: Representative weekly NO_2 profile with mean and ± 1 standard deviation.

Representative hour-of-day and day-of-week summaries for NO_2 are shown in Figure 8 and Figure 9. These summaries showed that pollutant concentrations did not remain constant across the day or week, and that variability also changed systematically across temporal regimes

Table 2: Summary of hours and weekdays variability for each pollutant.

Variability/Pollutants	NO_2	SO_2	$\text{PM}_{2.5}$	CO
Lowest-variability hour	5	14	23	5
Lowest std	9.65	1.20	6.69	0.20
Highest-variability hour	10	10	8	8
Highest std	14.23	3.82	12.63	0.44
Lowest-variability weekday	Sunday	Tuesday	Sunday	Sunday
Lowest std	12.07	2.35	9.18	0.29
Highest-variability weekday	Tuesday	Saturday	Monday	Tuesday
Highest std	14.80	2.98	11.11	0.36

The temporal variability results further showed pollutant-specific differences in when variability was highest and lowest. As summarised in Table 2, NO₂ showed its lowest hourly variability at 05:00 and highest hourly variability at 10:00, with the lowest weekday variability on Sunday and the highest on Tuesday. SO₂ showed its lowest hourly variability at 14:00 and highest at 10:00, with the lowest weekday variability on Tuesday and the highest on Saturday. PM_{2.5} showed its lowest hourly variability at 23:00 and highest at 08:00, with the lowest weekday variability on Sunday and the highest on Monday. CO showed its lowest hourly variability at 05:00 and highest at 08:00, with the lowest weekday variability on Sunday and the highest on Tuesday.

These findings indicate that pollutant behaviour is temporally structured not only at the mean level but also in its variability. This is important for forecasting because prediction difficulty may depend not only on forecast horizon but also on the temporal regime to which a forecast lead corresponds. The temporal analysis, therefore, provided an exploratory basis for both the interpretation of later models and the construction of temporal predictors.

4.4. Feature Engineering

Feature engineering was used to convert the key temporal and meteorological patterns identified in the exploratory analysis into model-ready predictors. In particular, the EDA showed meaningful associations between pollutant concentrations and meteorological variables, as well as a clear hourly and weekly temporal structure in the pollutant series. Based on these findings, the feature-engineering step focused on three main sources of predictive information: temporal cycles, wind-related effects, and recent lagged or rolling behaviour in pollutant and meteorological variables.

First, cyclical temporal variables were derived from the timestamp, including hour-of-day, day-of-week, and day-of-year. Because these variables are periodic, they were encoded using *sine* and *cosine* transforms rather than in raw numeric form to preserve circular continuity. For a cyclical variable x_t with period P , the encoding was defined as:

$$x_{sin,t} = \sin\left(\frac{2\pi x_t}{P}\right), \quad x_{cos,t} = \cos\left(\frac{2\pi x_t}{P}\right) \quad 4.1$$

Similarly, hour-of-day encoding was computed with $P = 24$, day-of-week encoding with $P = 7$, and day-of-year encoding with $P = 365.25$. Wind direction θ_t was transformed in a similar manner:

$$wd_{sin,t} = \sin(\theta_t), \quad wd_{cos,t} = \cos(\theta_t) \quad 4.2$$

And corresponding wind vector components were derived as

$$wind_{u,t} = ws_t \cdot wd_{cos,t}, \quad wind_{v,t} = ws_t \cdot wd_{sin,t} \quad 4.3$$

Where ws_t , indicates the speed of the wind. By using these modifications, directional effects were represented in a way that was appropriate for statistical and machine-learning models while maintaining cyclic continuity. This prevented false discontinuities, such as the change from Sunday to Monday or from hour 23 to hour 0. As direct predictors of recurrent daily, weekly, and weather-driven patterns, these temporal and wind encodings were particularly significant for the GBM and hybrid model.

Additional feature construction, including lagged and rolling predictors, was implemented selectively within individual modelling frameworks where required. This ensured that feature design remained consistent with the assumptions and structure of each model rather than being applied uniformly across all approaches.

4.5. Data Modelling

4.5.1. Forecasting Framework overview

The modelling framework also clearly distinguished between preprocessing completed once before model development and additional model-specific preparation carried out within each rolling window. The selected one-year dataset was preprocessed at the dataset level once, as described in Section 4.2, including the main imputation and outlier-handling pipeline used to create the cleaned modelling dataset. During rolling forecasting, additional train-window-only handling was applied only where required. These additional procedures included residual missing-value handling, training-slice standardization, past-data-only feature construction, and the preparation of forecast-like meteorological inputs. This distinction was necessary to preserve temporal validity and avoid using information from the prediction horizon during model fitting.

Within this shared framework, the models differ in how they represent pollutant dynamics. SARIMA and Holt–Winters use only the historical values of a single pollutant and therefore act as univariate baselines. VAR jointly models pollutants, meteorological variables, wind-related features, and temporal encodings within a multivariate linear system. GBM uses an explicitly engineered set of predictors to learn nonlinear relationships among pollutant history, meteorology, wind, and temporal structure. The hybrid (SARIMA-GBM) framework extends this further by first forecasting meteorological variables and then using those forecasts as drivers for pollutant prediction. In this way, the forecasting framework supports a structured comparison across different modelling families under the same experimental conditions.

4.5.2. Rolling Origin Evaluation and Window Configuration

All forecasting models were assessed using the same rolling-origin design to ensure a fair comparison across modelling techniques. For each forecast run, the model was trained on the most recent six months of hourly data and then used to generate forecasts for the next 72 hours. The forecast origin was then advanced by 72 hours, and the process was repeated throughout the chosen one-year study period. This architecture ensured that performance differences could be attributed to the modelling structure rather than to differences in data partitioning or testing methods, and it provided a uniform evaluation framework for all models.

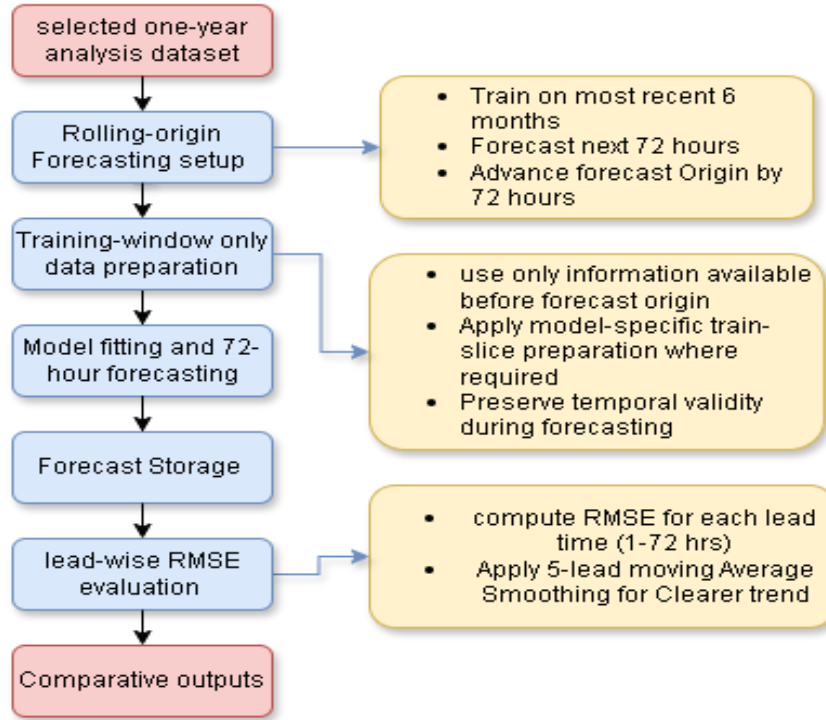


Figure 10: Common rolling-origin forecasting and evaluation structure used for all models.

The common forecasting logic applied to all models is summarised in Figure 10. The models were assessed using the same rolling-origin, forecast horizon, forecast storage format, and lead-wise RMSE calculation, despite differences in their internal structures and predictor designs.

Let t_k denote the k -th forecast origin, L the training-window length, and H the forecast horizon. For each forecast origin t_k , the training window is defined as:

$$T_k = [t_k - L, t_k - 1] \quad 4.4$$

Where, L corresponds to six months of hourly observations. The associated forecast horizon is defined as:

$$H_k = [t_k, t_k + H - 1] \quad 4.5$$

where $H=72$ hours. This notation makes explicit that each forecast is generated using only information available prior to the forecast origin.

Various forecast horizons could have been used, but the decision to use a 72-hour forecast horizon was primarily methodological. Because it offers a balanced short-to medium-term forecasting window, 72 hours was chosen for this study. It is long enough to assess forecast-skill degradation across a range of lead times and short enough to remain relevant for near-term warning and decision support. A matched 72-hour forecast step was used to create non-overlapping evaluation periods and a uniform basis for model comparison. Accordingly, the 72-hour update cycle was primarily a research-design choice rather than Quito's operational forecasting schedule.

Model development was limited to data available prior to the forecast origin for each rolling forecast windows. In practical terms, this means that all model fitting and additional model-time preparation relied solely on the corresponding training slice and not on observations from

the forecast horizon. This restriction was necessary to prevent data leakage, which would otherwise lead to artificially optimistic forecast performance.

As a result, extra training-window-only preparation was used during rolling forecasting when needed for a particular model, which is described in Section 4.5.1. The entire preparation pipeline outlined in Section 4.2 was not intended to be repeated within each rolling window. Instead, these rolling-window procedures served as model-time safeguards, ensuring that only pre-origin information was used for any remaining window-specific preparation.

Forecasts for leads $h = 1, 2, \dots, 72$ were then generated and saved along with pollutant identity, forecast timestamp, rolling-window index, and lead time. Because this rolling-origin architecture assesses predicting performance repeatedly over time rather than under a single fixed partition, it was selected over a single static train-test split. Because of this, the model comparison provides a more reliable basis for evaluating forecasting behaviour over the chosen year-long research window, as it is based on a wider range of meteorological and pollutant regimes.

4.5.3. Common Notation and Evaluation Metric

Throughout this study, a standard notation and evaluation process were employed to ensure uniformity across all forecasting models. Let p denote a target pollutant, and let $y_t^{(p)}$ represent the observed hourly concentration of pollutant p at time t . For a forecast generated at origin t_k , the prediction for lead time h is denoted by

$$\hat{y}_{t_k+h}^{(p)}, \quad h = 1, 2, \dots, 72 \quad 4.6$$

Where $h = 1$ denotes the first forecast hour following the forecast origin and $h = 72$ denotes the last hour in the 72-hour forecast horizon. To record and compare predictions from various rolling windows and models in a standard format, this notation was applied to all model outputs.

Forecast accuracy was evaluated using Root Mean Squared Error (RMSE). For a set of n forecast–observation pairs, RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad 4.7$$

where y_i represents the observed value, $\hat{y}_i$ represents the predicted value, and n denotes the total number of forecast–observation pairs. This metric is commonly employed in continuous air-quality forecasting because it penalises large forecast errors more strongly than small ones. To investigate how prediction accuracy varied over the 72-hour period, RMSE was computed independently for each forecast lead time rather than as a single overall value.

For pollutant p and forecast lead h , the lead-wise RMSE was computed as:

$$RMSE^{(p)}(h) = \sqrt{\frac{1}{n_h} \sum_{i=1}^{n_h} (y_{i,h}^{(p)} - \hat{y}_{i,h}^{(p)})^2} \quad 4.8$$

where n_h is the total number of forecast-observation pairs that are valid for lead time h after being aggregated over all rolling forecast windows. This approach allows for the comparison of both overall model accuracy and the rate of decline in forecast performance as the prediction horizon increases.

For visualisation purposes only, the RMSE curves were additionally subjected to a five-lead moving average. This smoothing was applied only to reduce short-scale visual fluctuation between neighbouring leads and to make the plotted trends easier to interpret. All reported RMSE values in tables, comparisons, and interpretations remained unsmoothed. In other words, smoothing was used only for graphical presentation and did not alter the evaluation metric itself.

This distinction is important because lead-wise instability may itself be informative. As a result, the raw RMSE values remain the basis for quantitative comparisons among models and pollutants, and the smoothed curves were utilised solely as visual aids.

4.6. Model-Specific Implementation

The common forecasting and evaluation framework used for all models was explained in Section 4.5. The implementation specifics for each modelling approach are now presented in this section. The discussion of each model emphasises its role within the comparison framework, the input data it employs, the formulation used in this study, and the primary procedures for rolling forecasts. Unless a model requires an extra step specific to its implementation, common components such as the rolling-origin design, forecast horizon, and lead-wise RMSE evaluation are not repeated in full.

Additional clarification on missing-value handling during rolling forecasting

Because leakage prevention was a central methodological concern, the treatment of residual missing values during rolling forecasting differed by model:

- SARIMA and Holt-Winters: only the training target series was imputed when residual missing values remained.
- VAR and direct GBM: training-fitted seasonal-median rules were applied to both the training slice and the forecast-slice predictors, where needed.
- Hybrid (SARIMA-GBM): future meteorological inputs were not directly imputed from future observations; instead, they were generated by Stage A meteorological forecasting, and any remaining forecast-feature gaps were handled using training-derived fallback statistics only.

In the rolling-window imputation rules used for the study, the seasonal grouping was defined explicitly as:

$$g(t) = (\text{day} - \text{of} - \text{week}(t), \quad \text{hour} - \text{of} - \text{day}(t)) \quad 4.9$$

That is, the seasonal group corresponds to the combination of day-of-week and hour-of-day. When seasonal-median imputation was required within a rolling window, the training-group median for group g was computed as:

$$m_g = \text{median}\{y_t: t \in T_k, \quad g(t) = g\} \quad 4.10$$

where m_g denotes the training-slice median for seasonal group g . If a required seasonal group had insufficient usable observations, the implementation fell back to interpolation or training-slice median filling, depending on the model and data structure.

4.6.1. Univariate Seasonal Time-Series Forecasting Using SARIMA

The main univariate statistical baseline for pollution forecasting in the study was SARIMA. Its purpose was to determine the degree of predictive skill that could be derived from a single pollutant's past behaviour without using meteorological variables, cross-pollutant data, or externally constructed predictors. Because of this, SARIMA can serve as a benchmark model for comparing more complex multivariate, machine-learning, and hybrid techniques.



Figure 11: SARIMA forecasting workflow within one rolling forecast window.

The SARIMA forecasting process for a single pollutant within a single rolling forecast window is shown in Figure 11. Residual invalid or incomplete values were examined and dealt with utilising training-window-only rules following the extraction of the pollutant training slice. After that, a fixed seasonal SARIMA specification was applied, with the option to switch to a more straightforward non-seasonal ARIMA model if necessary. The 72-hour prediction for that pollutant was ultimately produced using the fitted model.

Each pollutant was modelled independently using only its own hourly concentration series. Let, $y_t^{(p)}$ represent the pollutant p observed concentration at time t . The SARIMA model was fitted only to the pollutant training slice at each rolling forecast origin before forecasting the following 72 hours. As a result, the model's input was limited to the univariate training series.

$$\{y_t^{(p)}: t \in T_k\} \quad 4.11$$

where T_k is the training window described in Section 4.5.2. Instead of using an explicit calendar covariate, seasonal behaviour was internally reflected by the model structure.

The generic SARIMA form is written as:

$$SARIMA(p, d, q) \times (P, D, Q)_s \quad 4.12$$

where p, d, q denote the non-seasonal autoregressive, differencing, and moving-average orders, P, D, Q denote the corresponding seasonal orders, and s is the seasonal period. Using the backshift operator B , the model may be written as:

$$\Phi(B^s) \phi(B) (1 - B)^d (1 - B^s)^D y_t = \theta(B^s) \theta(B) \varepsilon_t \quad 4.13$$

where $\phi(B)$ and $\theta(B)$ are the non-seasonal autoregressive and moving-average polynomials, $\Phi(B^s)$ and $\Theta(B^s)$ are the seasonal autoregressive and moving-average polynomials, and ε_t is a white-noise error term. In this formulation, the use of $\Theta(B^s)$ for the seasonal moving-average term distinguishes it clearly from the non-seasonal moving-average polynomial $\theta(B)$.

In this study, SARIMA specification was fixed as:

$$SARIMA(1, 0, 1) \times (1, 0, 1)_{24} \quad 4.14$$

utilising an intercept term. To represent daily seasonality in hourly air pollution data, the seasonal period $s = 24$ was selected. This specification was used as a fixed benchmark configuration for comparison forecasting rather than being chosen independently for each pollutant or rolling timeframe via AIC, BIC, or grid search. The earlier Augmented Dickey-Fuller results, which showed that the cleaned one-year series were stationary within the selected research window, validated the choice of $d = D = 0$. As a result, SARIMA was employed here as a deliberately chosen seasonal benchmark, enabling the contribution of univariate temporal structure to be evaluated uniformly across pollutants and directly contrasted with the study's more intricate models.

Before fitting, negative pollutant concentrations in the training slice were treated as invalid and converted to missing values. Any residual incomplete values were then handled using the training-window-only seasonal-median rules described above. For SARIMA, imputation was applied only to the training target series, and no forecast-window target values were imputed. No forecast-window target values were imputed.

The seasonal SARIMA model was fitted first for each rolling window. The implementation switched back to a more straightforward non-seasonal ARIMA (1,0,1) specification if this fit failed due to numerical instability or local estimation issues. To ensure that forecasts could still be generated for troublesome windows and that isolated fitting failures would not disrupt the entire rolling evaluation, this fall-back was incorporated. Following fitting, the model produced projections for the following 72 hours, which were then saved in the standard long-format structure required for lead-wise RMSE assessment.

SARIMA fulfils two related functions within the larger comparative framework. In the first place, it offers an open standard for the predictive power of temporal structure, particularly to

pollutants alone. Second, it provides the statistical forecasting element that is subsequently utilized in the hybrid (SARIMA-GBM) architecture for forecasting weather. In this way, SARIMA serves as a methodological building element within the larger modelling system as well as a stand-alone univariate baseline.

4.6.2. Univariate Seasonal Smoothing with Holt-Winters

Holt–Winters was employed in this study as the exponential-smoothing counterpart to SARIMA within the univariate forecasting group. Instead of using autoregressive and moving-average dynamics, its function was to produce a seasonal baseline based on recursive smoothing of level, trend, and seasonal components. This enables Holt-Winters, which uses a conceptually distinct statistical method from SARIMA, to serve as a complementary benchmark for assessing the predictive skill that can be derived from a pollutant's own repeating temporal structure.

Each pollutant was modelled separately using only its own hourly concentration series, much like in the SARIMA scenario. The Holt-Winters model was fitted to the pollutant training slice at each rolling origin, and forecasts for the following 72 hours were produced. Therefore, weather variables, cross-pollutant data, and externally manufactured predictors were excluded from the model input, which was limited to the univariate pollutant series during the training window.

The additive Holt–Winters formulation was adopted because the cleaned pollutant series exhibited recurring daily cycles with approximately stable seasonal amplitude over the selected one-year study window. Let y_t denote the pollutant concentration at time t . The additive Holt–Winters model maintains three recursively updated latent components: level l_t , trend b_t , and seasonal component s_t . These are updated as:

$$\begin{aligned} l_t &= \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1}) \\ b_t &= \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1} \\ s_t &= \gamma(y_t - l_t) + (1 - \gamma)s_{t-m} \end{aligned} \tag{4.15}$$

where α , β , and γ are smoothing parameters and m is the seasonal period. Next, the h -step-ahead projection is provided by

$$\hat{y}_{t+h} = l_t + hb_t + s_{t-m+h} \tag{4.16}$$

To reflect daily seasonality in hourly pollutant concentrations, the seasonal period in this study was set at $m = 24$. The smoothing parameters were automatically computed within each training window rather than being predetermined, and an additive trend and seasonal components were employed.

The Holt–Winters forecasting procedure for a single pollutant within a single rolling forecast window is summarised in Figure 12.

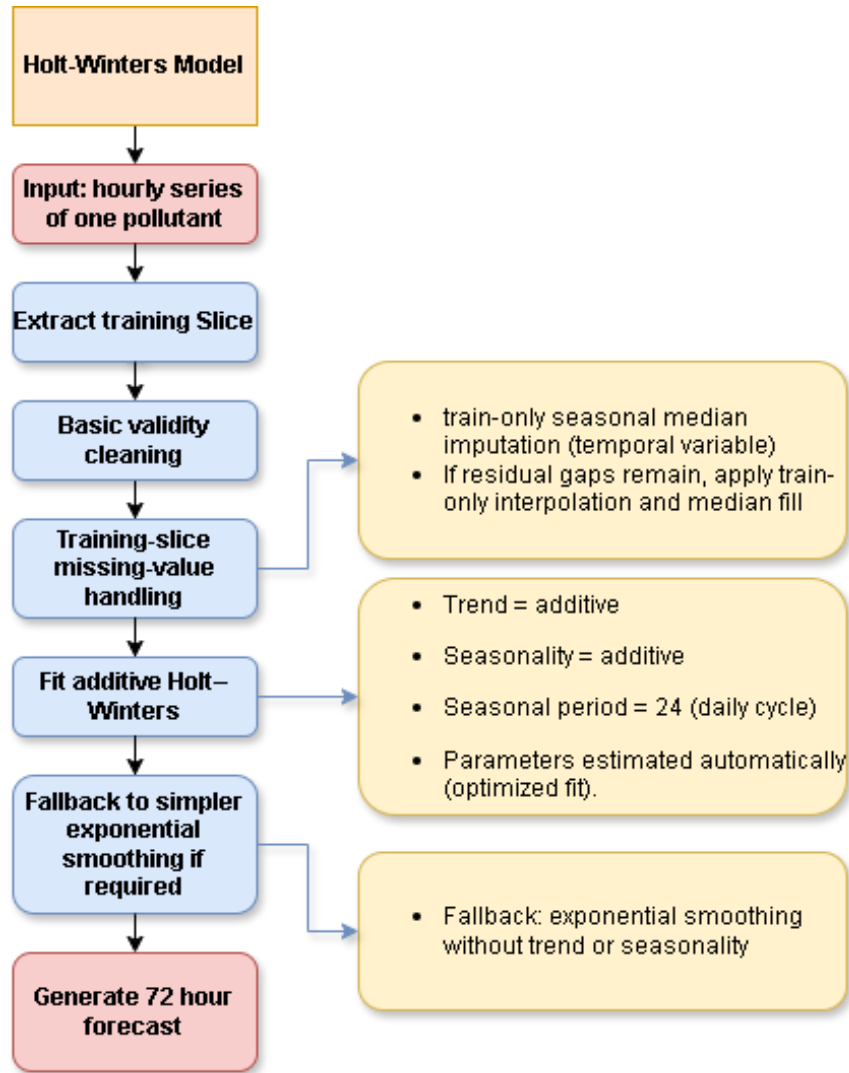


Figure 12: Holt-Winters forecasting workflow within one rolling forecast window.

Before fitting, negative pollutant concentrations in the training slice were treated as invalid and converted to missing values. Any residual incomplete values were then handled using the same training-window-only seasonal-median rules described above. As with SARIMA, only the training target series was imputed; no forecast-window target values were imputed.

The additive Holt-Winters model was initially fitted to each rolling window using automatic parameter estimation and daily seasonality with a period of 24. The method switched back to a more straightforward exponential smoothing configuration without trend or seasonality if the seasonal fit failed due to numerical instability or inadequate variation. Instead of letting the fitting process fail in nearly all cases, this fallback provides a reliable baseline forecast. Following fitting, the model generated forecasts for the following 72 hours, which were then saved in the standard long-format structure for lead-wise RMSE assessment.

In addition to SARIMA, Holt-Winters serves as a second univariate benchmark in this study's comparison. Holt-Winters assesses the predictive value of recursively smoothed level, trend, and seasonal behaviour, while SARIMA measures the predictive value of autoregressive seasonal structure. Before switching to multivariate, machine-learning, and hybrid techniques, it is helpful to compare these two baselines to determine the relative contribution of autoregressive dynamics against smoothing-based seasonal regularity to short-term forecasting skill.

4.6.3. Multivariate Forecasting Using Vector Autoregression (VAR)

The primary multivariate linear benchmark in the study was VAR. Its goal was to determine whether modelling pollutant concentrations alongside meteorological factors, wind-related transformations, and temporal encodings as opposed to relying solely on a pollutant's history improved forecasting performance. VAR thus symbolises the shift from univariate statistical baselines to a multivariate system in which several variables change simultaneously over time in the comparative framework of this study.

Within each rolling forecast window, a single multivariate system was fitted in the implemented VAR model. Pollutant concentrations, weather parameters, wind-related encodings, and temporal cyclical encodings, as described in Section 4.4, comprised the endogenous vector at time t , denoted by y_t . In particular, the meteorological group contained temperature, rainfall, solar radiation, relative humidity, air pressure, and wind speed, whereas the pollutant group comprised NO₂, SO₂, PM_{2.5}, and CO. Consequently, the endogenous system that has been implemented can be expressed as:

$$y_t = \begin{bmatrix} p_t \\ m_t \\ w_t \\ c_t \end{bmatrix} \quad 4.17$$

where p_t denotes pollutant concentrations, m_t meteorological variables, w_t wind-related encodings and derived components, and c_t temporal cyclical encodings.

The VAR model was fitted in its standard linear multivariate form,

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \dots + A_p y_{t-p} + \varepsilon_t \quad 4.18$$

where c is a constant vector, $A_1, \dots, A_p$ are coefficient matrices, p is the lag order, and ε_t is a multivariate innovation term. Under this formulation, each variable in the system depends on its own lagged values and on the lagged values of all other included variables.

The VAR forecasting procedure used within a single rolling forecast window is summarised in Figure 13.

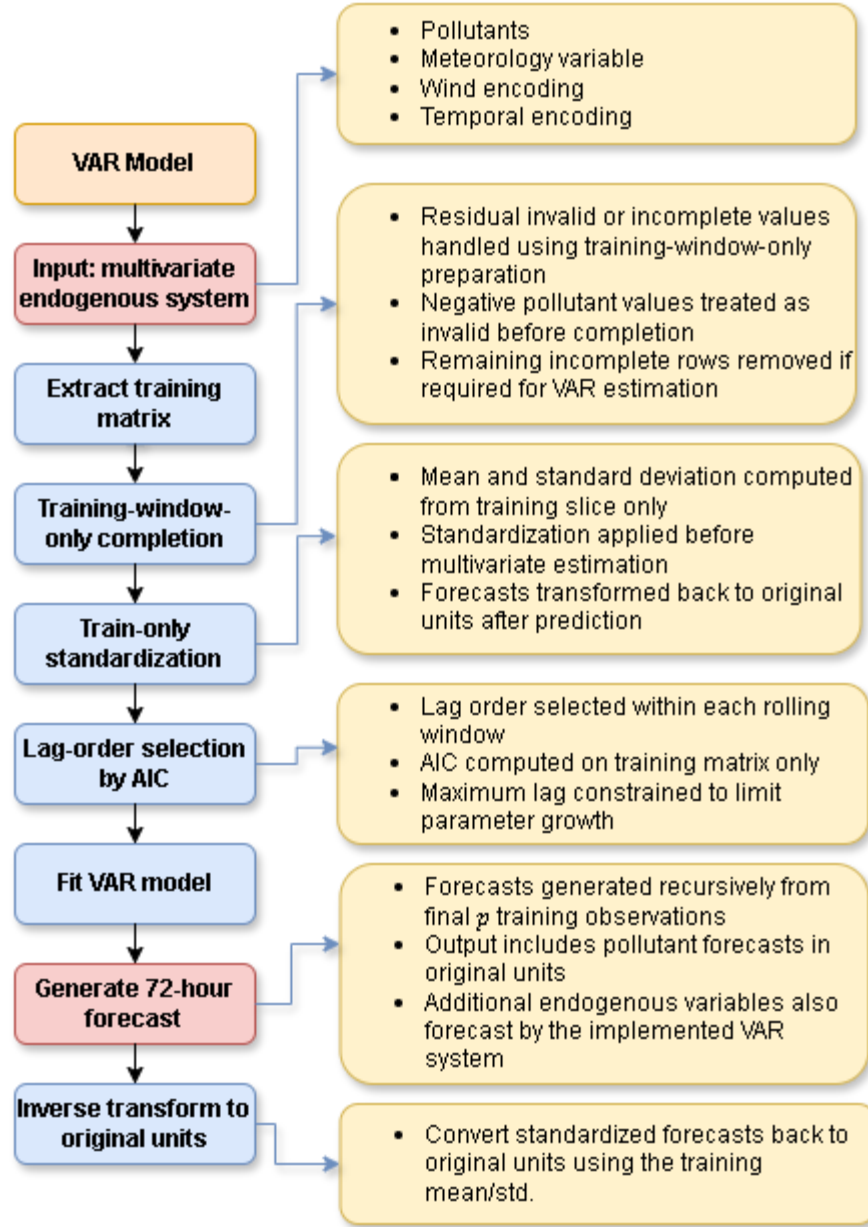


Figure 13: VAR forecasting workflow within one rolling forecast window.

The training-window-only method outlined in Section 4.5.2 was used to create the multivariate training matrix prior to estimation. The mean and standard deviation of the training slice alone were used to normalize all variables in the VAR system after residual invalid or incomplete values had been addressed where necessary. For a variable $z_{t,j}$ the standardized form was:

$$z_{t,j}^{(std)} = \frac{Z_{t,j} - \mu_j}{\sigma_j} \quad 4.19$$

where μ_j and σ_j are the training-slice mean and standard deviation for variable j . This step was included to reduce numerical instability, since the variables in the system were measured on different scales. After forecasting, the predicted standardized values were transformed back to the original units using the inverse relation.

$$\hat{z}_{t,j} = \hat{z}_{t,j}^{(std)} \sigma_j + \mu_j \quad 4.20$$

So that the results remained interpretable in physical concentration and meteorological units.

Using the Akaike Information Criterion (AIC) calculated only on the training matrix, the lag order p was chosen independently inside each rolling forecast window. Lag search was limited by a maximum lag to prevent excessive parameter expansion in a high-dimensional system. This decreased the possibility of unstable estimation while enabling the model to adjust its lag structure to the available training data. If lag-order selection failed, or if the selected VAR (p) specification could not be fitted because of numerical or estimation issues, the implementation fell back to a simpler VAR (1) model.

Following the selection of the lag order and the fitting of the VAR model, forecasts were produced iteratively for the following 72 hours using the final p observations from the standardized training matrix as initial conditions. As a result, the fitted system generated forecasts for the extra meteorological, wind, and temporal encoding variables that were part of the endogenous vector in addition to pollutants.

Instead of being treated as deterministic exogenous regressors, the temporal cyclical encodings and derived wind components were included in the endogenous system of the implemented VAR model. Consequently, these quantities were forecast by the fitted model, along with the pollutant and meteorological variables. Conceptually, a VARX or VARMAX formulation with exogenous regressors would be required for a more rigorous treatment of deterministic calendar encodings. Therefore, the current VAR model should not be interpreted as a formally specified deterministic-regressor VARX model, but rather as an empirical multivariate benchmark based on the implemented conventional VAR system.

The VAR model was designed to evaluate the extent to which a simple linear multivariate structure could enhance forecasting in comparison to the univariate baselines within the broader comparative framework. Before switching to more adaptable nonlinear techniques such as GBM and the hybrid (SARIMA-GBM) framework, it is useful in this study to assess whether cross-variable dependence provides valuable predictive information.

4.6.4. Nonlinear Forecasting Using Gradient Boosting Machine (GBM)

The primary nonlinear machine-learning benchmark in this study was GBM. Its purpose was to determine whether forecasting ability would be enhanced by using an explicitly constructed predictor set that combined meteorological data, temporal structure, wind-related changes, and pollutant history to predict pollutant concentrations. Therefore, GBM reflects the shift from statistical time-series models to a flexible nonlinear regression approach that may learn interactions that are challenging to characterise through univariate or linear multivariate formulations in the comparative context of this study.

The GBM technique did not rely exclusively on the historical values of a single pollutant, unlike SARIMA and Holt-Winters. Rather, a structured set of predictors comprising exogenous and history-based variables was used to train a separate GBM model for each target pollutant. A weekend indicator, wind-related encoding, cyclical temporal encodings, and meteorological factors were among the exogenous predictors. To maintain cycle continuity and prevent false discontinuities, time variables and wind direction were represented using sine and cosine encodings, as explained in Section 4.4. These predictors were added to enable the model to respond to directional transport effects, meteorological forcing, and recurrent daily and weekly patterns.

The GBM models employed synthetic features obtained from pollution history in addition to these exogenous predictors. Lagged predictors were developed at horizons for every pollutant p .

$$L \in \{1, 2, 3, 6, 12, 24, 48, 72\} \quad 4.21$$

So that a lag feature can be written as:

$$x_t^{(p,L)} = y_{t-L}^{(p)} \quad 4.22$$

Additionally, rolling summary statistics were created to depict recent regime behaviour. The rolling mean and rolling standard deviation for a rolling window of length w were defined as:

$$\begin{aligned} \mu_t^{(p,w)} &= \frac{1}{w} \sum_{j=1}^w y_{t-j}^{(p)} \\ \sigma_t^{(p,w)} &= \sqrt{\frac{1}{w} \sum_{j=1}^w \left(y_{t-j}^{(p)} - \mu_t^{(p,w)} \right)^2} \end{aligned} \quad 4.23$$

These rolling features were calculated for 24-hour and 72-hour intervals in the implemented model. Each pollutant model has the ability to incorporate not only its own historical behaviour but also cross-pollutant dependence and recent variability in a form appropriate for nonlinear prediction because a distinct GBM model was trained for each target pollutant while the predictor matrix contained lagged and rolling information from all pollutants.

The GBM model can be written generically as:

$$\hat{y}(x) = \sum_{m=1}^M v f_m(x) \quad 4.24$$

Where, x is the predictor vector, $f_m(\cdot)$ is the m^{th} regression tree, M is the number of boosting stages (trees), v is the learning rate controlling the contribution of each new tree. With each new tree fitted to lower the residual prediction error, the forecast function is constructed progressively as an additive ensemble of weak learners under this formulation. Because each pollutant in this study had its own gradient boosting regressor, the nonlinear predictor-response relationship could vary by pollutant type.

The GBM forecasting procedure used within a single rolling forecast window is summarised in Figure 14.

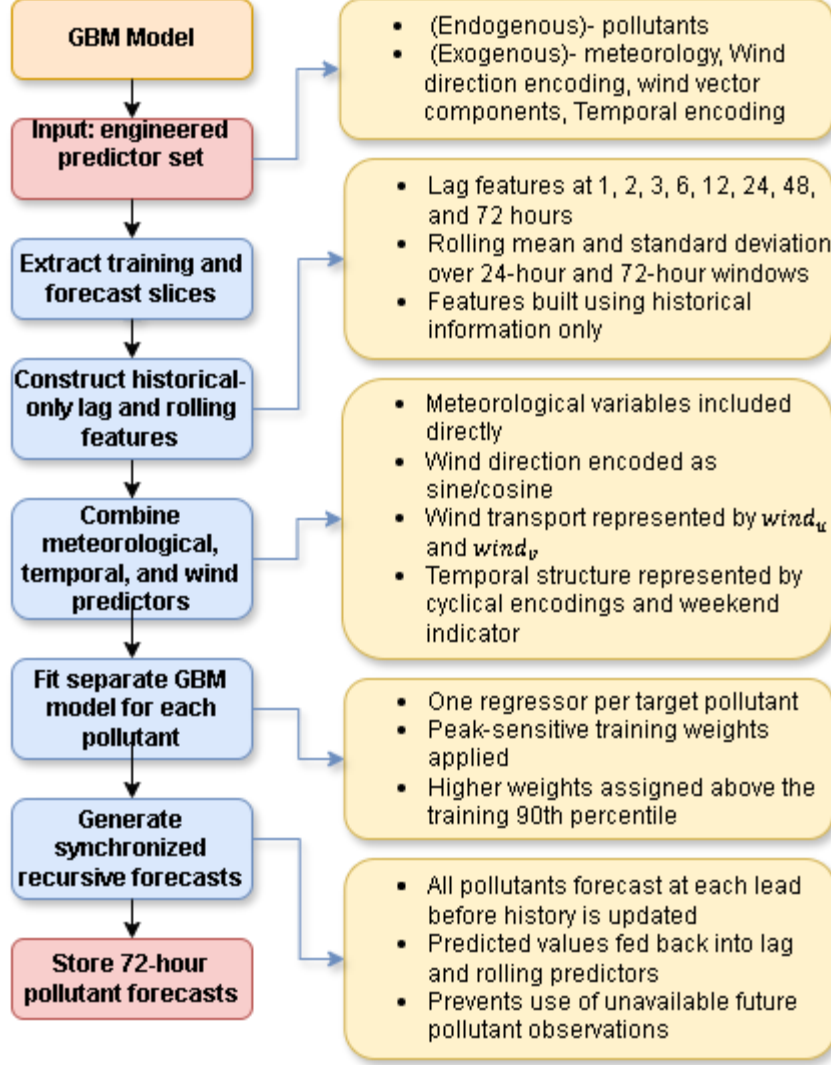


Figure 14: GBM forecasting workflow within one rolling forecast window.

The distinction between predictors readily available at forecast time and those requiring recursive updating was a crucial aspect of the GBM implementation. The forecast slice in the direct GBM architecture provided exogenous inputs for the forecast horizon, including wind-related properties, temporal encodings, and meteorological variables. On the other hand, rolling summaries for future leads and pollutant lags had to be rebuilt recursively because they were not directly available. The implementation employed a synchronised recursive forecasting technique to overcome this. The pollutant-specific forecast at each forecast lead t can be expressed as:

$$\hat{y}_t^{(p)} = f_p(x_t^{exo}, x_t^{endo}) \quad 4.24$$

Where $f_p(\cdot)$ represents the GBM regressor for pollutant p , x_t^{exo} indicates the exogenous predictor set and x_t^{endo} represents the rolling and lagging predictors based on the history of pollutants. Prior to building the lagged and rolling predictors for the subsequent step, all pollutants were predicted for a particular forecast step, and the newly projected values were added to the history table. Instead of using observations that were not accessible throughout the prediction horizon, this synchronised recursive process ensured that future pollutant attributes were derived from previously projected values.

Peak-sensitive weighting was also used in the training of the pollutant-specific GBM models $q_{0.90}^{(p)}$ denote the 90th percentile of the training target distribution for pollutant p . The sample weight for observation i was then defined as:

$$w_i = \begin{cases} w_{peak} & y_i^{(p)} > q_{0.90}^{(p)} \\ w_{base}, & otherwise \end{cases} \quad 4.25$$

Where, $w_{peak} > w_{base}$. Baseline observations were given weight 1 in the implemented model, while observations that exceeded the training 90th percentile were given weight 3. Because elevated pollution episodes are less common but more significant for operational forecasting, this weighting scheme was chosen. The model was encouraged to maintain higher sensitivity to pollution peaks when using the entire training sample by boosting the influence of higher-concentration cases during training.

GBM is the primary nonlinear benchmark in this study's comparison framework. It is useful for determining whether a sufficiently rich, well-designed predictor set can enhance forecasting performance beyond the linear multivariate VAR system and univariate statistical baselines. GBM offers the study a flexible machine-learning reference against which the hybrid (SARIMA-GBM) framework can subsequently be compared because it integrates meteorology, temporal encodings, wind-related transformations, pollutant lags, and rolling summaries within a recursive multi-step forecasting design.

4.6.5. Hybrid Weather-First Forecasting using (SARIMA-GBM)

The most operationally limited forecasting model in this investigation was the hybrid (SARIMA-GBM) framework. Its goal was to assess pollution prediction under the practical assumption that future meteorological data would not be directly available at forecast time. The hybrid design first anticipates meteorological factors and then uses those projected drivers to predict pollution concentrations, in contrast to the direct GBM model, which uses observed meteorological inputs from the forecast slice. Therefore, in the comparative framework of this study, the hybrid model represents a weather-first forecasting pipeline designed to assess the degree of predictive accuracy that may be achieved when future weather must be estimated rather than presumed to be known.

A central distinction between the direct GBM framework and the hybrid weather-first framework lies in how meteorological information is handled at forecast time. In the direct GBM model, observed meteorological values from the forecast slice are available as predictors. In the hybrid framework, by contrast, future meteorological observations are not assumed to be known. Instead, they are first forecasted in Stage A and then used as inputs for pollutant prediction in Stage B. Consequently, the hybrid design is more consistent with operational forecasting conditions, but it also introduces a second layer of uncertainty, since any error in the meteorological forecast stage may propagate into the subsequent pollutant forecast stage.

Within each rolling forecast frame, the hybrid architecture was executed in two successive steps. In Stage A, forecast-like meteorological inputs were produced by fitting distinct univariate time-series models to the meteorological variables. These forecast-like meteorological factors were combined with deterministic temporal predictors to train and apply pollution models in Stage B. The absence of pollutant lags and pollutant rolling summaries as predictors in the hybrid pollutant stage is a crucial aspect of this implementation's design. Because pollutant prediction is based on predicted meteorological and temporal structure rather than recursive pollutant-history predictors, the hybrid model is essentially different from the Direct GBM approach.

Let m_t denote the vector of meteorological variables and wind-related transformations, and let $y_t^{(p)}$ denote pollutant p . The hybrid framework may be written schematically as:

4.26

$$\hat{m}_{t:t+H-1} = S(m_{T_k})$$

$$\hat{y}_{t:t+H-1}^{(p)} = f_p(\hat{m}_{t:t+H-1}, \phi(\hat{m}_{t:t+H-1}), c_{t:t+H-1})$$

In which $S(\cdot)$ represents the meteorological forecasting stage, $f_p(\cdot)$ represents the pollutant model for pollutant p , $\phi(\cdot)$ represents engineered meteorological predictors derived from forecast like meteorological stream, c_t represents the temporal encodings that are deterministic. Instead of using direct pollutant-history features, this formulation bases the pollution forecast on time-based predictors and projected weather.

The hybrid forecasting procedure used within a single rolling forecast window is summarised in Figure 15.

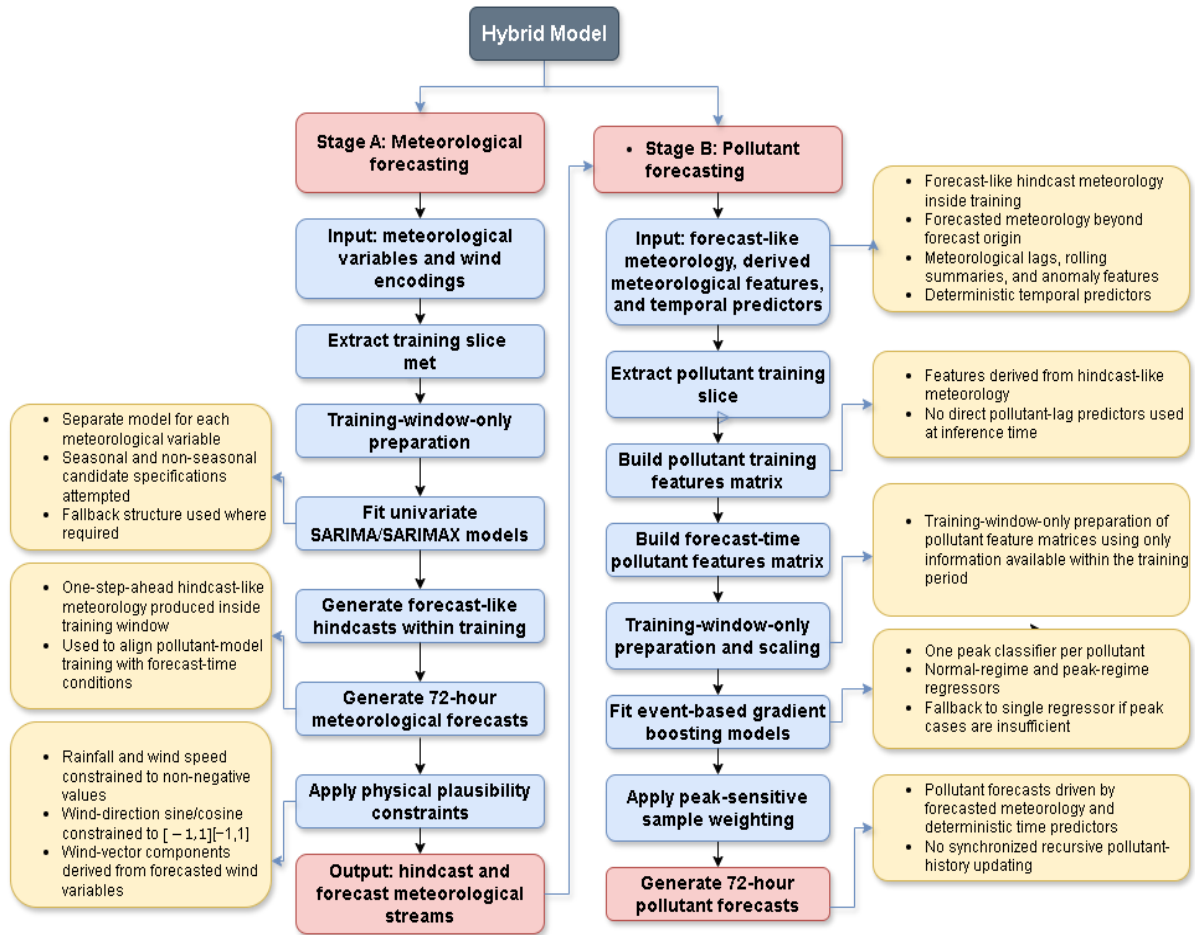


Figure 15: Hybrid (SARIMA-GBM) forecasting workflow within one rolling forecast window.

Stage A: Meteorology Forecasting

In order to generate the meteorological inputs needed for pollution prediction, Stage A was created. For every meteorological series, temperature, precipitation, solar radiation, relative humidity, air pressure, wind speed, and the sine and cosine components of wind direction separate univariate SARIMA/SARIMAX-type models were fitted. Then, instead of being directly forecasted, wind-vector components were constructed from forecasted wind-speed and forecasted wind-direction encodings.

During the meteorological forecasting stage, a number of candidate specifications were tried to increase resilience. These contained a non-seasonal fall-back in addition to seasonal and differentiated seasonal variants. Practically speaking, the candidate set was made up of a daily seasonal $(1,0,1) \times (1,0,1)_{24}$ form, a seasonal differenced $(1,1,1) \times (1,0,1)_{24}$ form, as well as a more straightforward non-seasonal $(1,0,1) (1,0,1)$ fall-back. This candidate structure was chosen to offer a stable and somewhat flexible meteorological forecasting stage across variables and rolling windows rather than as a comprehensive model search.

The employment of forecast-like hindcasts within the training window was a crucial methodological aspect of the hybrid implementation. Perfectly observed meteorological was not used to train the pollution model. Rather, the downstream pollutant model learned from inputs that resembled the uncertainty structure it would encounter at inference time by using the fitted meteorological models to generate one-step-ahead filtered or forecast-like meteorological estimates within the training slice. The goal of this alignment was to promote consistency between training and deployment settings and lessen optimism bias.

Stage A generated a complete 72-hour weather forecast horizon following fitting. Then, when necessary, physical plausibility limitations were implemented. For instance, sine and cosine components were limited to staying inside $[-1, 1]$, while wind speed and rainfall were clipped at zero. The predicted wind speed and wind-direction encodings were then used to calculate the resulting wind-vector components.

Stage B: Pollutant Forecasting from Forecasted Meteorology

Stage B predicted pollution concentrations using the forecast-like meteorological stream generated in Stage A. Using a machine-learning architecture based on histogram-based gradient boosting, a distinct pollutant model was built for each pollutant. Each pollutant's predictor matrix was made up of three parts:

- Base meteorological predictors: calculated wind-vector components and forecast-like meteorological variables
- Lagged meteorological values and rolling summaries obtained from the forecast-like meteorological stream are examples of engineered meteorological features.
- The temporal encodings previously described in Section 4.4 are deterministic temporal predictors.

For meteorological variable m_t , lagged predictors were generated at horizons:

$$L \in \{1, 3, 6, 12, 24\} \quad 4.27$$

So that a lag feature may be written as:

$$x_t^{(m,L)} = m_{t-L} \quad 4.28$$

Additionally, rolling mean and rolling standard deviation were also calculated over 24-hour and 72-hours windows to characterize recent atmospheric regimes:

$$\mu_t^{(m,w)} = \frac{1}{w} \sum_{j=1}^w m_{t-j} \quad 4.29$$

$$\sigma_t^{(m,w)} = \sqrt{\frac{1}{w} \sum_{j=1}^w (m_{t-j} - \mu_t^{(m,w)})^2}$$

Furthermore, anomaly-style predictors were developed to depict deviations from current weather patterns. An anomalous term for a given rolling window w can be expressed as:

$$\alpha_t^{(m,w)} = m_t - \mu_{t-1}^{(m,w)} \quad 4.30$$

The purpose of these built meteorological predictors was to enable the pollutant model to react to changes in atmospheric circumstances and recent meteorological regime behaviour in addition to the present predicted weather.

The pollutant models were trained on forecast-like meteorological inputs instead of directly observed meteorology, which was an important implementation decision. To put it another way, rather than using the directly observed training meteorology, the meteorological predictors employed during training were the hindcast outputs from Stage A. The purpose of this design was to expose the pollutant model to the same imprecise meteorological data it would encounter during forecasting.

Events-Based Pollutant Modelling

To increase sensitivity to high pollution levels, the pollutant stage employed an event-based design. The model included the following for each pollutant:

- A peak classifier,
- A normal-regime regressor,
- and a peak-regime regressor

The 90th percentile of the pollutant training distribution was used to define the peak regime. Let p_t indicate the expected likelihood that time t falls within the peak regime, and allow $\hat{y}_t^{(norm)}$ and $\hat{y}_t^{(peak)}$ indicate the normal-regime and peak-regime forecasts. Next, a probability-weighted combination was created to create the final forecast:

$$\hat{y}_t = (1 - \alpha p_t) \hat{y}_t^{(norm)} + (\alpha p_t) \hat{y}_t^{(peak)} \quad 4.31$$

Where α is the scaling factor that is applied to the event-based combination rule's peak probability. In this study, the scaling factor was set to alpha equals 0.85. The purpose of this structure was to enable the model to react differently in both typical and elevated pollution regimes.

Peak-sensitive weighting was also used during hybrid pollutant-model training. In the code, observations above the 90th percentile received a larger peak weight than in the direct GBM stage; specifically, the peak weight was set to 6, while the baseline weight remained 1. This was done because elevated pollution episodes, although less common, are operationally important

Scaling and Forecast Generation

After the pollutant models were fitted, the hybrid framework generated a 72-hour pollutant forecast horizon directly from the 72-hour anticipated meteorological stream. Since the hybrid pollutant stage did not use pollutant-history predictors, the synchronized recursive pollutant-updating strategy used in the direct GBM model was not required. Because Stage B

immediately maps meteorological forecasts to pollution estimates once Stage A creates the forecasts, the hybrid architecture is simpler to employ during inference time.

4.7. Summary of Models and Input Information

The forecasting models utilized in this study are compiled in Table 3. along with the primary categories of input data that each model uses. While all models were assessed using the same rolling-origin paradigm outlined in Section 4.5.2, there were significant differences in their input structures, modelling assumptions, and prediction processes. It is crucial to make these distinctions clear so that performance comparisons can be understood in terms of modelling design rather than variations in the evaluation process.

Table 3: Comparison of models on utilization of pollutant and meteorological information.

Model	Model Type	Pollutant History Used	Meteorology Used
SARIMA	Univariate statistical time-series	Yes (own pollutant series only)	No
Holt–Winters	Univariate exponential smoothing	Yes (own pollutant series only)	No
VAR (classic)	Multivariate linear statistical	Yes (all pollutants jointly via VAR system)	Yes (meteorology, wind encodings, temporal variables as endogenous)
GBM (direct)	Machine learning (nonlinear, feature-based)	Yes (lagged and rolling features of pollutants; cross-pollutant history)	Yes (observed meteorology + derived wind + temporal features)
Hybrid Weather-First (SARIMA → GBM)	Hybrid statistical–machine learning (weather-First)	No direct pollutant lags (pollutant history not used as predictors)	Yes (forecasted meteorology from SARIMA + temporal features)

5. Results and Analysis

5.1. Comparative Results Overview

This section presents the comparative forecasting results across the 1–72-hour prediction horizon. Performance is assessed using lead-wise RMSE under a consistent rolling-origin evaluation setting described in Section 4.5.

The results should be interpreted as a controlled comparison of model behaviour under a shared preprocessing and evaluation framework, rather than as an exhaustive hyperparameter optimization exercise. While all models were evaluated using identical data partitions and forecasting conditions, the tuning effort was not equalized across modelling approaches.

Table 4 summarises RMSE by pollutant and forecast window segment. The 72-hour horizon was divided into three segments: 1–24 h, 25–48 h, and 49–72 h. This segmentation was used

only to summarise how model performance changes across the horizon; it does not redefine the overall forecasting range.

Table 4: Best model by pollutant and forecast window based on lead-wise RMSE.

Pollutant	Horizon	SARIMA	VAR	Holt-Winters	GBM	HYBRID	Best model
CO (mg/m ³)	(1–24 h)	0.228	0.238	0.241	0.183	0.233	GBM
	(25–48 h)	0.290	0.286	0.332	0.240	0.292	GBM
	(49–72 h)	0.284	0.289	0.339	0.269	0.289	GBM
	Overall (1–72 h)	0.267	0.271	0.304	0.231	0.271	GBM
NO ₂ (µg/m ³)	(1–24 h)	8.70	8.23	10.75	7.21	7.77	GBM
	(25–48 h)	10.76	10.71	12.92	8.77	9.99	GBM
	(49–72 h)	10.98	11.11	13.93	9.06	10.38	GBM
	Overall (1–72 h)	10.15	10.02	12.53	8.35	9.38	GBM
PM _{2.5} (µg/m ³)	(1–24 h)	6.52	6.39	7.57	6.56	6.95	VAR
	(25–48 h)	7.79	7.74	8.94	8.23	8.59	VAR
	(49–72 h)	7.31	7.49	8.86	8.17	8.26	SARIMA
	Overall (1–72 h)	7.21	7.21	8.46	7.65	7.93	VAR
SO ₂ (µg/m ³)	(1–24 h)	1.341	1.348	1.579	1.479	1.338	HYBRID
	(25–48 h)	1.657	1.679	1.867	1.979	1.618	HYBRID
	(49–72 h)	1.705	1.756	1.844	2.170	1.665	HYBRID
	Overall (1–72 h)	1.568	1.594	1.763	1.876	1.540	HYBRID

Note: RMSE values are compared only within each pollutant, since pollutant scales and units differ. Raw RMSE magnitudes are therefore not comparable across pollutants.

Figures 16–19 complement Table 4 by showing how forecast error evolves across the individual lead times. These figures are computed from the full set of valid forecast-observation pairs aggregated across all rolling forecast windows for each pollutant, model, and lead time, rather than from a single forecast run.

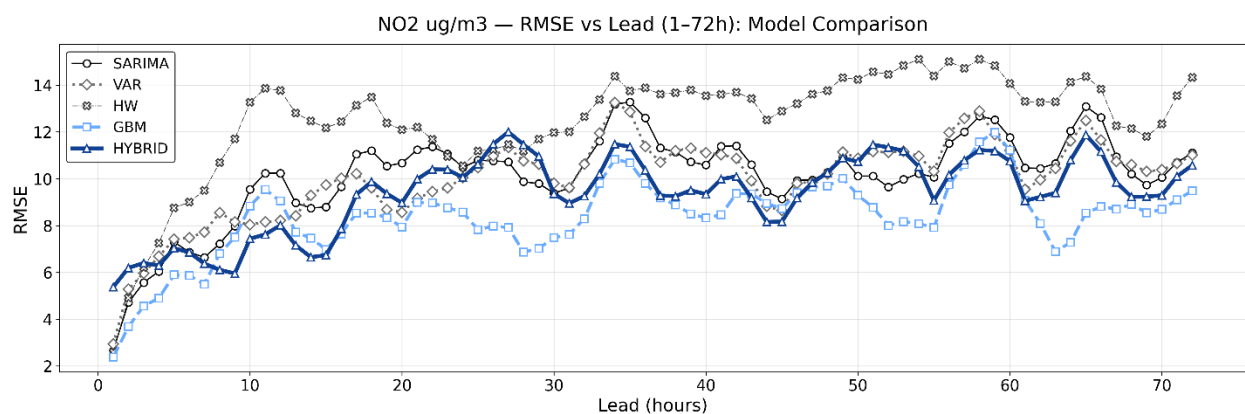


Figure 16: Lead-wise RMSE for NO₂ across the five forecasting models.

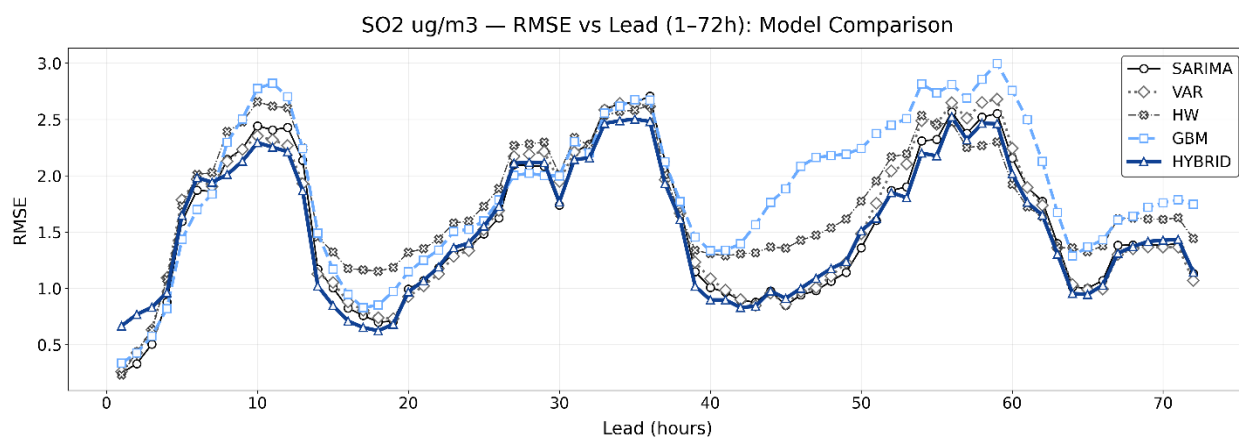


Figure 17: Lead-wise RMSE for SO_2 across the five forecasting models.

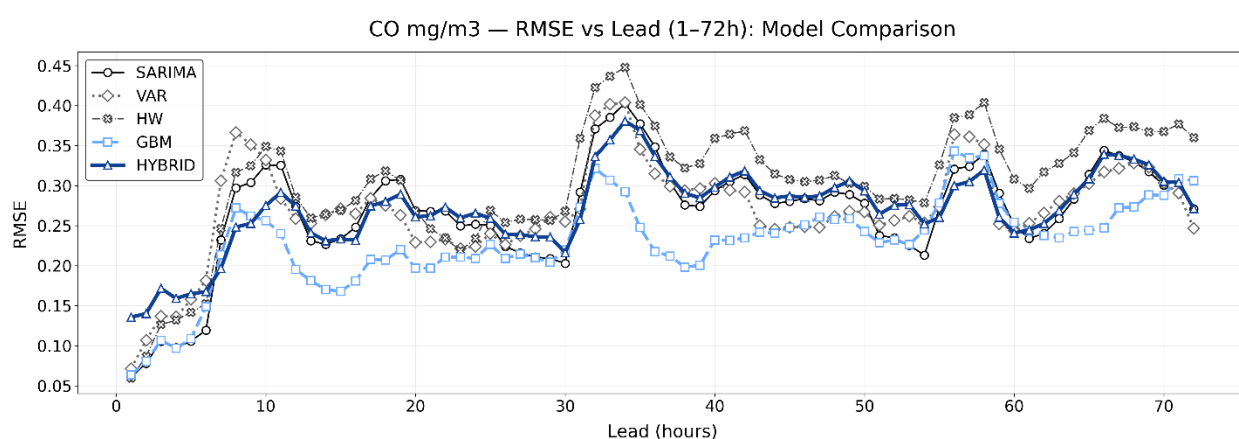


Figure 18: Lead-wise RMSE for CO across the five forecasting models.

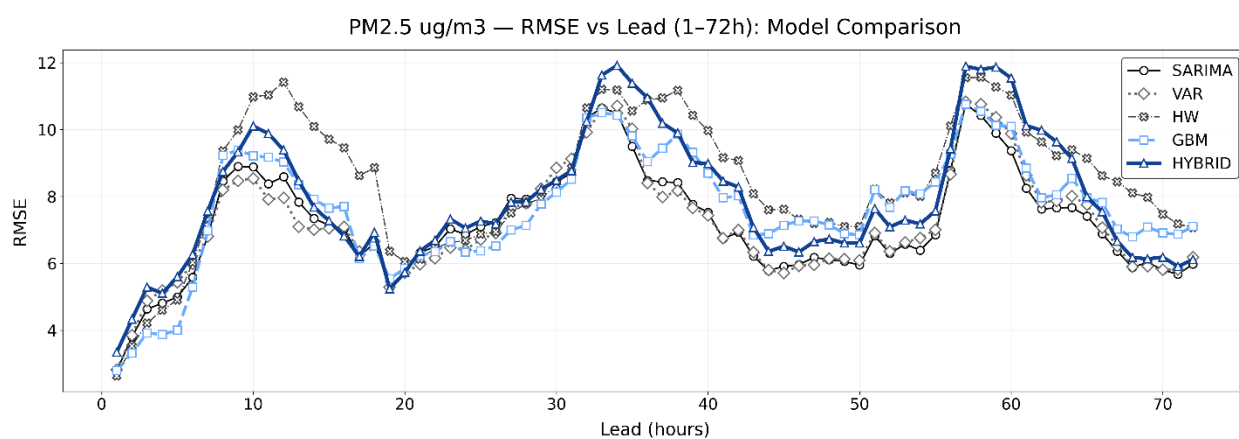


Figure 19: Lead-wise RMSE for $\text{PM}_{2.5}$ across the five forecasting models.

Overall, the results show that no single model was best for all pollutants across the full 72-hour horizon. Model performance differed by pollutant, and in the case of $\text{PM}_{2.5}$, also by forecast-window segment.

5.2. Pollutant Specific Comparative Results

This section reports the main pollutant-specific findings from Table 4 and figure 16-19. The focus is on model rankings and forecast behavior, while deeper interpretation of why these patterns occurred is reserved for discussion section.

5.2.1. Nitrogen Dioxide (NO₂)

For NO₂, GBM produced the lowest RMSE in all three forecast-window segments and also achieved the lowest overall RMSE across the full 1–72-hour horizon. VAR was the closest competing model in the early and middle parts of the horizon, while SARIMA performed better than Holt–Winters across all segments. The hybrid model remained competitive but did not outperform direct GBM.

The lead-wise RMSE curve for NO₂ showed non-monotonic variation across the horizon, with recurring changes in error magnitude. However, the ranking of the best-performing model remained stable across the three forecast-window segments.

5.2.2. Sulfur Dioxide (SO₂)

For SO₂, the hybrid (SARIMA-GBM) framework produced the lowest RMSE in all three forecast-window segments and also achieved the lowest overall RMSE. SARIMA and VAR remained relatively close to the hybrid model at earlier leads, while GBM and Holt–Winters showed larger errors over the longer horizon segments.

Compared with the other pollutants, the SO₂ model differences were narrower and the lead-wise RMSE curves appeared smoother. The best-performing model remained stable across the full 72-hour horizon.

5.2.3. Carbon Monoxide (CO)

For CO, GBM produced the lowest RMSE in each forecast-window segment and also achieved the lowest overall RMSE. VAR remained close to SARIMA and the hybrid model in some segments, while Holt–Winters produced the highest overall error among the five models.

Because CO is measured in mg/m³, while NO₂, SO₂, and PM_{2.5} are measured in µg/m³, the RMSE values for CO should be interpreted only within the CO comparison. They should not be compared directly in magnitude with the RMSE values of the other pollutants.

The CO lead-wise RMSE curves showed variation across lead time, but the ranking of the strongest model remained stable across the forecast-window segments.

5.2.4. Fine Particulate Matter (PM_{2.5})

PM_{2.5} showed the clearest change in model ranking across the forecast horizon. VAR produced the lowest RMSE in the 1–24 h and 25–48 h segments, while SARIMA produced the lowest RMSE in the 49–72 h segment. In the overall 1–72 h comparison, VAR and SARIMA were very close after rounding.

GBM and the hybrid model remained competitive but did not provide the lowest RMSE in any of the three forecast-window segments. Holt–Winters showed the highest overall RMSE for PM_{2.5}.

The PM_{2.5} results show that relying only on an overall RMSE value can hide changes in model ranking across the forecast horizon.

5.3. Forecast-Horizon Behaviour Across the 1–72-Hours Window

The lead-wise RMSE curves show that forecast accuracy changes across the 1–72-hour horizon. Although error generally increased at later leads, the increase was not strictly monotonic. Instead, the curves contained recurring peaks and troughs, indicating that forecasting difficulty varied within the forecast window.

This pattern was visible across the pollutants, but with different strength. The purpose of this section is not to restate the best-performing model for each pollutant, but to show that lead-wise evaluation provides information that an overall RMSE value would not capture. In particular, the lead-wise curves show whether performance remains stable across the horizon or changes between earlier and later forecast leads.

The oscillatory behaviour is consistent with the temporal variability identified during exploratory analysis, where pollutant variability differed by hour-of-day and day-of-week. However, no separate quantitative test was conducted to directly link hourly variability with lead-wise RMSE variation. Therefore, this explanation should be interpreted as qualitative and evidence-consistent rather than formally verified.

Overall, the horizon-wise analysis shows that the 72-hour forecast window should not be treated as a single uniform block. Lead-wise evaluation is necessary because both error magnitude and model stability can vary across the forecast horizon.

5.4. Synthesis of Results

Overall, the results show that short- to medium-term air-quality forecasting in Quito is pollutant-specific and partly horizon-sensitive. The strongest model was not universal across pollutants, and the lead-wise analysis showed that performance can vary within the 72-hour forecast window.

The findings support the primary hypothesis in a qualified way. Models using meteorological information, multivariate structure, or nonlinear learning often improved performance relative to the univariate baselines, but their advantage was conditional rather than universal.

6. Discussion

6.1. Implications of Forecast-Horizon Behaviour

The horizon-wise patterns indicate that forecast uncertainty is structured rather than simply increasing with distance from the forecast origin. In a 72-hour forecast, later leads naturally carry more uncertainty because models rely on information that becomes progressively less current. However, the observed lead-wise behaviour suggests that this accumulation of uncertainty interacts with recurring temporal conditions in the pollutant series.

A plausible explanation is that some forecast leads repeatedly coincide with periods of higher atmospheric or emission variability, such as specific hours-of-day or parts of the weekly cycle. During these periods, pollutant concentrations may change more abruptly because of traffic activity, boundary-layer dynamics, wind conditions, or other short-term meteorological effects. Forecasts corresponding to these regimes are therefore more difficult than forecasts aligned with more stable periods. This interpretation is consistent with the exploratory analysis,

although it remains qualitative because no separate statistical test was conducted to directly link temporal variability with lead-wise RMSE.

This has an important methodological implication. Evaluating only one overall RMSE value can hide within-horizon instability and may give the impression that model performance is uniform across the full forecast window. Lead-wise evaluation is therefore necessary because it shows not only average accuracy, but also how reliable a forecast remains at different points within the 1-72-hour horizon. For operational use, this means that confidence in a forecast should be understood as lead-dependent rather than constant across the entire prediction window.

6.2. Why Model Performance Differed Across Pollutants

The differences in model performance can be understood through the interaction between model assumptions and pollutant behaviour. Each modelling family represents the air-quality process differently. Therefore, model performance depends not only on the complexity of the model, but also on whether its assumptions match the dominant behaviour of the pollutant being forecast.

The univariate models rely mainly on each pollutant's own past values. This makes them useful when persistence and regular daily structure are strong. Their performance shows that hourly pollutant series contain meaningful autocorrelation and daily structure. However, these models cannot directly represent meteorological forcing or cross-pollutant dependence. This limits their usefulness when pollutant concentrations are strongly affected by changing dispersion conditions, wind behaviour, humidity, or interactions with other pollutants. This finding is consistent with earlier air-quality forecasting literature, where univariate models are often useful baselines but are usually limited when meteorological variability becomes important.

The multivariate linear model adds cross-variable information, but it still assumes that relationships are mostly linear and additive. This can be useful when pollutant and meteorological variables move together in a relatively stable way. However, it is less suitable for threshold effects, sudden changes, or nonlinear interactions between emissions, dispersion, and meteorology. This helps explain why multivariate structure was helpful in some cases but did not provide a universal advantage.

The nonlinear feature-based model is more flexible because it can combine pollutant history, meteorological variables, wind features, and temporal encodings without requiring a fixed linear relationship. This makes it better suited to pollutants whose behaviour is shaped by persistence together with nonlinear meteorology-emission interactions. This interpretation is consistent with the literature reviewed earlier, where machine-learning models are often useful when pollutant dynamics depend on richer predictor interactions.

The hybrid weather-first framework represents a different assumption. It does not simply add more predictors; it changes how future meteorological information is handled. Instead of assuming future observed weather is available, it first forecasts meteorological variables and then uses those forecasted drivers for pollutant prediction. This design is more realistic for operational forecasting, but it also introduces uncertainty from the meteorological forecast stage. Therefore, the hybrid model is most useful when the benefit of explicitly modelling future atmospheric conditions outweighs the error introduced by forecasting those conditions.

Pollutant characteristics help explain why no single model dominated across all cases. CO and NO₂ are closely linked to combustion and traffic-related activity, so their short-term behaviour may contain strong persistence combined with nonlinear effects from meteorology and hour-of-day patterns. SO₂ appeared more suitable for the weather-first structure,

suggesting stronger dependence on atmospheric conditions in the selected dataset. PM_{2.5} is more complex because it can reflect both direct emissions and secondary formation, as well as accumulation and dispersion processes. This mixed behaviour helps explain why its most suitable model structure changed across the forecast horizon.

Overall, the results suggest that model complexity is beneficial only when it matches the structure of the pollutant process. Nonlinear learning, multivariate dependence, and weather-first forecasting each added value under different conditions, but none was universally superior. This is why the primary hypothesis is supported in a qualified sense: richer modelling frameworks often improved performance, but their benefit depended on pollutant behaviour and forecast lead.

6.3. Practical Implications for Forecasting in Quito

The practical implication is that air-quality forecasting in Quito should be pollutant-specific and horizon-aware. The findings do not support using one model family automatically for every pollutant. Instead, model choice should consider the pollutant's temporal structure, its sensitivity to meteorological conditions, and the forecast lead range that matters most for the intended application.

For pollutants where temporal persistence, emission timing, and meteorological conditions interact in complex ways, flexible nonlinear models may provide the most useful forecasting structure. Where pollutant behaviour is strongly tied to atmospheric evolution, a weather-first approach may be preferable, provided that the meteorological forecasts are reliable. For pollutants with mixed formation and accumulation processes, model selection may need to vary by forecast horizon rather than relying on a single overall ranking.

The results also imply that forecast uncertainty should not be communicated as constant across the 72-hour window. Some forecast leads may be less reliable because they align with more variable temporal regimes or because prediction uncertainty accumulates over time. A practical forecasting system would therefore benefit from reporting not only predicted pollutant levels, but also lead-specific confidence or reliability information.

However, the present study should be interpreted as a controlled comparative assessment rather than as a finished operational forecasting system. Operational deployment would require actual numerical weather forecasts, validation over additional years, monitoring for temporal drift, and testing across changing emission and meteorological conditions. The study provides guidance for model selection, but further work is needed before these results can be translated directly into real-time forecasting practice.

6.4. Study Limitations

Several limitations should be considered when interpreting the findings. These limitations do not invalidate the comparison, but they define the conditions under which the results should be understood.

First, the analysis was limited to one selected year of hourly data. This allowed a controlled rolling-origin comparison and captured daily, weekly, and within-year seasonal variation. However, it does not show whether the same model behaviour would remain stable across multiple years, unusual pollution episodes, or different meteorological regimes.

Second, the study used data from a single monitoring station. This helped isolate temporal forecasting behaviour, but it did not capture spatial transport, regional background pollution,

or differences between monitoring locations. The results are therefore strongest as evidence for local temporal forecasting rather than city-wide spatial forecasting.

Third, the hybrid framework used forecast-like meteorological inputs generated within the study rather than real operational numerical weather forecasts. As a result, the hybrid findings show how the weather-first design behaved under the study conditions, but not necessarily how it would perform with real forecast products. In addition, the propagation of meteorological forecast error into pollutant forecast error was not quantified through a separate sensitivity analysis.

Fourth, multi-step forecasting allows uncertainty to accumulate across the forecast horizon. This is especially relevant for models that use recursive prediction or forecasted drivers. Later-horizon errors therefore reflect both pollutant complexity and the cumulative difficulty of multi-step forecasting.

Fifth, each modelling family has structural limitations. SARIMA and Holt–Winters are transparent but univariate. VAR includes multivariate relationships but assumes linearity. The VAR implementation treated deterministic temporal encodings as endogenous variables; a VARX or VARMAX formulation would provide a more theoretically rigorous treatment in future work. GBM and the hybrid framework are more flexible, but they depend more strongly on feature engineering, input quality, and careful validation. The results should therefore be interpreted as reflecting a trade-off between interpretability, flexibility, and operational realism.

Finally, the findings should not be generalized directly to all pollutants or all urban settings. Similarly, Quito's topography, meteorology, and emissions context may produce forecasting behaviour that differs from other cities.

Taken together, these limitations show that the study provides a strong comparative foundation for short- to medium-term air-quality forecasting in Quito, but not a final deployment-ready system. Its main contribution is to show that model suitability depends on pollutant behaviour and forecast horizon under a controlled, leakage-aware evaluation design.

7. Conclusion

This study set out to evaluate the effectiveness of different modelling approaches for short- to medium-term air-quality forecasting in Quito using a consistent rolling-origin framework. The results show that forecasting performance cannot be reduced to a single best model. Instead, model suitability depends on both the pollutant being predicted and, in some cases, the position within the forecast horizon.

The findings demonstrate that incorporating additional structure beyond univariate temporal patterns can improve forecasting performance, but only when that structure aligns with the behaviour of the pollutant. Multivariate relationships, nonlinear interactions, and meteorological drivers each contributed useful information under specific conditions, but no modelling approach was universally superior across all pollutants and forecast leads.

A key contribution of this study is the emphasis on horizon-aware evaluation. The lead-wise analysis showed that forecast accuracy and reliability vary within the 72-hour window, and that these variations are structured rather than random. This highlights the importance of evaluating forecasting models not only in terms of overall accuracy, but also in terms of how performance evolves across the prediction horizon.

From a practical perspective, the results suggest that air-quality forecasting in Quito should be approached as a pollutant-specific and horizon-sensitive task. Model selection should consider both the characteristics of the pollutant and the intended forecast lead time, rather

than relying on a single model for all conditions. In addition, forecast uncertainty should be interpreted as varying across the prediction window rather than being constant.

The study also provides a controlled comparative framework that can be extended in future work. Further research could include multi-year validation, incorporation of real numerical weather prediction data, exploration of direct multi-horizon forecasting methods, and extension to pollutants with more complex formation processes such as ozone.

In summary, the study shows that short- to medium-term air-quality forecasting is inherently conditional. Model performance depends on the interaction between pollutant dynamics, modelling assumptions, and forecast horizon. Recognising this conditionality is essential for both methodological development and practical forecasting applications.

8. Future Work

Several directions can extend the present study. First, the analysis should be repeated using a longer temporal record to test whether the comparative model rankings remain stable across years and under more extreme meteorological or pollution conditions. Second, future work should include multiple monitoring stations to represent spatial dependence and pollutant transport more explicitly, rather than relying on a single-station temporal framework.

Third, the hybrid framework should be evaluated using real operational meteorological forecasts rather than forecast-like statistical weather inputs. This would provide a more realistic test of how meteorological forecast uncertainty affects downstream pollutant prediction. Fourth, future studies could compare the recursive forecasting design used here with direct multi-horizon forecasting strategies to examine whether error accumulation at later leads can be reduced.

Finally, further work should extend the framework in two important ways: by improving model interpretability, especially for machine-learning and hybrid approaches, and by considering additional pollutant types, particularly secondary pollutants such as ozone. Together, these directions would help move the present study from a controlled comparative framework toward a broader and more operationally robust forecasting system for urban air quality.

References

- Abdul Rahim, U., & Masseran, N. (2021). Visualization of multivariate time series pollutant variables in Malaysia. *Journal of Physics: Conference Series*, 1988(1). <https://doi.org/10.1088/1742-6596/1988/1/012089>
- Ali, D., Singh, M., Khan, A. A., Vidyarthi, A. K., & Ahmed, S. (2025). Impact of meteorological drivers on air quality index: a case study from Delhi. *Environmental Pollution*, 388, 127324. <https://doi.org/10.1016/j.envpol.2025.127324>
- Alleyne, G., Breman, J. G., Claeson, M., Evans, D. B., Jamison, D. T., Jha, P., Measham, A. R., Mills, A., & Musgrove, P. (2006). *Disease control priorities in developing countries, second edition* (Vol. 1).
- Bai, L., Wang, J., Ma, X., & Lu, H. (2018). Air pollution forecasts: An overview. *International Journal of Environmental Research and Public Health*, 15(4), 780. <https://doi.org/10.3390/ijerph15040780>
- Baklanov, A., & Zhang, Y. (2020). Advances in air quality modeling and forecasting. *Global Transitions*, 2, 261–270. <https://doi.org/10.1016/j.glt.2020.11.001>
- Balasubramanian, A. (2022). Improving air quality prediction using gradient boosting. *International Journal on Science and Technology (IJSAT)*, 13, 1–9. <https://doi.org/10.5281/zenodo.14631450>
- Bhatti, U. A., Yan, Y., Zhou, M., & Ali, S. (2021). Time series analysis and forecasting of air pollution particulate matter (PM_{2.5}): An SARIMA and factor analysis approach. *IEEE Access*, 9, 41019–41031. <https://doi.org/10.1109/ACCESS.2021.3060744>
- Cevallos, V. M., Díaz Suárez, M. V., & Sirois, C. S. (2017). Particulate matter air pollution from city of Quito, Ecuador, activates inflammatory signaling pathways in vitro. *Innate Immunity*, 23(4), 392–400. <https://doi.org/10.1177/1753425917699864>
- Chau, P. N., Zalakeviciute, R., Thomas, I., & Rybarczyk, Y. (2022). Deep learning approach for assessing air quality during COVID-19 lockdown in Quito. *Frontiers in Big Data*, 5 <https://doi.org/10.3389/fdata.2022.842455>
- Fetahi, E., Ovuoraye, P., Canhasi, E., Susuri, A., & Misini, A. (2025). Multi-horizon air pollution prediction using interpretable machine learning in a growing urban area. *Ecological Engineering & Environmental Technology*, 26(8). <https://doi.org/10.12912/27197050/207283>
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: principles and practice* (3rd ed.). OTexts.
- Jayanth, T., Manimaran, A., Reddy, V. R. K., & Rajashekar, N. (2025). Enhancing air quality prediction through Holt–Winters Smoothing and Transformer-BiGRU with Bayesian Optimization. *IEEE Access*, 13. <https://doi.org/10.1109/ACCESS.2025.3621231>
- Kumar, A., & Goyal, P. (2011). Forecasting of daily air quality index in Delhi. *Science of The Total Environment*, 409(24), 5517–5523. <https://doi.org/10.1016/j.scitotenv.2011.08.069>
- Mahendra, H., Swamy, M., Kumar, D., Kumari, S., Kashyap, S., Fulwani, S., & Chatterjee, A. (2023). Assessment and prediction of air quality level using ARIMA model: a case study of Surat City, Gujarat state, India. *Nature Environment and Pollution Technology*, 22(1), 199–210. <https://doi.org/10.46488/NEPT.2023.v22i01.018>
- Malik, I., Ilyas, A., & Uduman, P. S. (2020). Forecasting ambient air quality of Chennai city In India. *Geography, Environment, Sustainability*, 13(3), 12–21. <https://doi.org/10.24057/2071-9388-2019-97>
- Rumbe, G., Hamasha, M. M., & Almashaqbeh, S. (2024). A comparison of Holt-Winters and Artificial Neural Network approach in forecasting: A case study for tent

- manufacturing industry. *Results in Engineering*, 21(1).
<https://doi.org/10.1016/j.rineng.2024.101899>
- Rybarczyk, Y. P., & Zalakeviciute, R. (2021). Assessing the COVID-19 impact on air quality: A Machine Learning Approach. *Geophysical Research Letters*, 48.
<https://doi.org/10.1029/2020GL091202>
- Samal, K. K. R., Babu, K. S., Das, S. K., & Acharaya, A. (2019). *Time Series based air pollution forecasting using SARIMA and Prophet model* In proceedings of the 2019 International Conference on Information Technology and Computer Communications, Singapore, Singapore. <https://doi.org/10.1145/3355402.3355417>
- Vlachokostas, C., Nastis, S. A., Achillas, C., Kalogeropoulos, K., Karmiris, I., Moussiopoulos, N., Chourdakis, E., Baniyas, G., & Limperi, N. (2010). Economic damages of ozone air pollution to crops using combined air quality and GIS modelling. *Atmospheric Environment*, 44(28), 3352–3361.
<https://doi.org/10.1016/j.atmosenv.2010.06.023>
- World Health Organization. (2024). Ambient (outdoor) air pollution.
[https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)
- Yi, M., & Lin, F. (2025). A Hybrid air quality prediction method based on VAR and Random Forest. *Journal of Computer and Communications*, 13(02), 142–154.
<https://doi.org/10.4236/jcc.2025.132009>
- Zahedi, A., Jaafarzadeh, N., Mir, I., & Tahmasebi Birgani, Y. (2024). Forecasting air quality index (AQI) and air quality health index (AQHI) by time series models in Ahvaz, Iran. *Environmental Health Engineering and Management*, 11(4), 469–476.
<https://doi.org/10.34172/EHEM.2024.46>
- Zhang, J., & Ding, W. (2017). Prediction of air pollutants concentration based on an extreme learning machine: The case of Hong Kong. *International Journal of Environmental Research and Public Health*, 14(2). <https://doi.org/10.3390/ijerph14020114>

Appendix

This appendix provides selected supporting diagnostics related to data preprocessing, stationarity assessment, and feature engineering. Only representative tables and figures are included to support key methodological steps, while avoiding duplication of patterns already presented in the main results.

A: Missing Data and Cleaning Summary

Table A.1: Missing value percentage before preprocessing for the selected one-year dataset.

Variable	Missing Count	% Missing
Temperature (°C)	63	0.717
Rain (mm)	356	4.053
Solar Radiation (W/m ²)	62	0.706
Relative Humidity (%)	63	0.717
Wind Speed (m/s)	66	0.751
Wind Direction (degrees)	66	0.751
Atmospheric Pressure (mb)	61	0.694
NO ₂ (µg/m ³)	258	2.937
SO ₂ (µg/m ³)	252	2.869
CO (mg/m ³)	604	6.876
O ₃ (µg/m ³)	376	4.281
PM _{2.5} (µg/m ³)	133	1.514

Table A.2: Percentage of detected outliers per variable

Variable	Outlier Percentage (%)
NO ₂ (µg/m ³)	0.638
SO ₂ (µg/m ³)	1.935
O ₃ (µg/m ³)	0.444
PM _{2.5} (µg/m ³)	0.467
CO (mg/m ³)	1.184
Temperature (°C)	0.000
Rain (mm)	1.332
Solar Radiation (W/m ²)	0.273
Relative Humidity (%)	0.000
Wind Speed (m/s)	0.342
Wind Direction (degrees)	0.000
Atmospheric Pressure (mb)	0.228

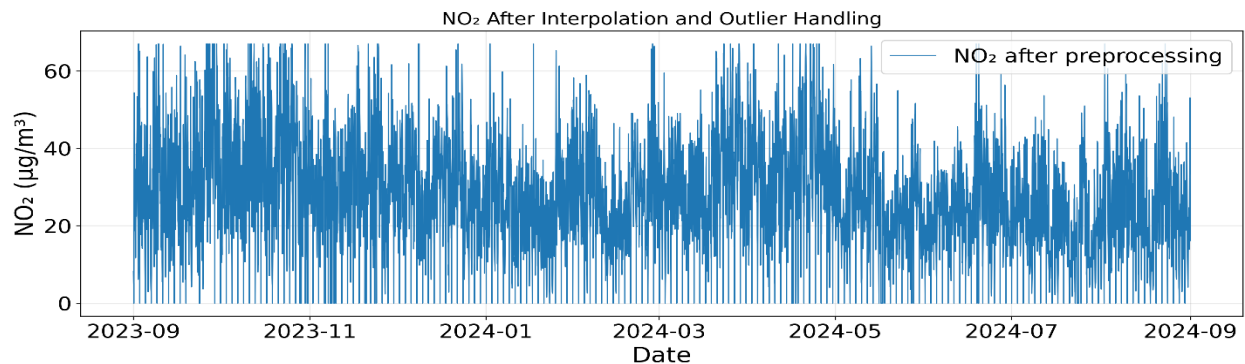


Figure A.1: Representative NO₂ time series after missing-value interpolation and outlier handling.

Tables A.1 and A.2 summarise the extent of missing values and the proportion of observations identified as outliers prior to preprocessing.

Figure A.1 provides a representative example of the NO₂ time series after preprocessing, illustrating the combined effect of missing-value interpolation and outlier treatment. The absence of gaps and the reduction of extreme spikes indicate the effectiveness of the applied preprocessing steps.

B: Stationarity test Results (ADF)

Note: p-values are reported as $p < 0.001$ where the numerical output rounded to 0.0000.

Table B.1: Augmented Dickey–Fuller (ADF) test results for pollutant and meteorological variables.

Variable	ADF Statistic	p-value	1% Critical	5% Critical	10% Critical	Stationarity
NO ₂ (µg/m ³)	-8.7879	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
SO ₂ (µg/m ³)	-12.8768	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
O ₃ (µg/m ³)	-6.7048	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
PM _{2.5} (µg/m ³)	-10.2519	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
CO (mg/m ³)	-8.7656	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
Temperature (°C)	-9.8211	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
Rain (mm)	-11.1008	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
Solar Radiation (W/m ²)	-11.1044	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
Relative Humidity (%)	-6.7153	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
Wind Speed (m/s)	-8.3559	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
Wind Direction (degrees)	-12.3206	< 0.001	-3.4311	-2.8619	-2.5669	Stationary
Atmospheric Pressure (mb)	-9.3211	< 0.001	-3.4311	-2.8619	-2.5669	Stationary

Table B.1 reports the Augmented Dickey–Fuller (ADF) test results for the pollutant and meteorological variables used in the study. For all variables, the unit-root null hypothesis was rejected at the 1% significance level, indicating that the cleaned time series can be treated as stationary over the selected one-year window.

C: Temporal and Seasonal Structure

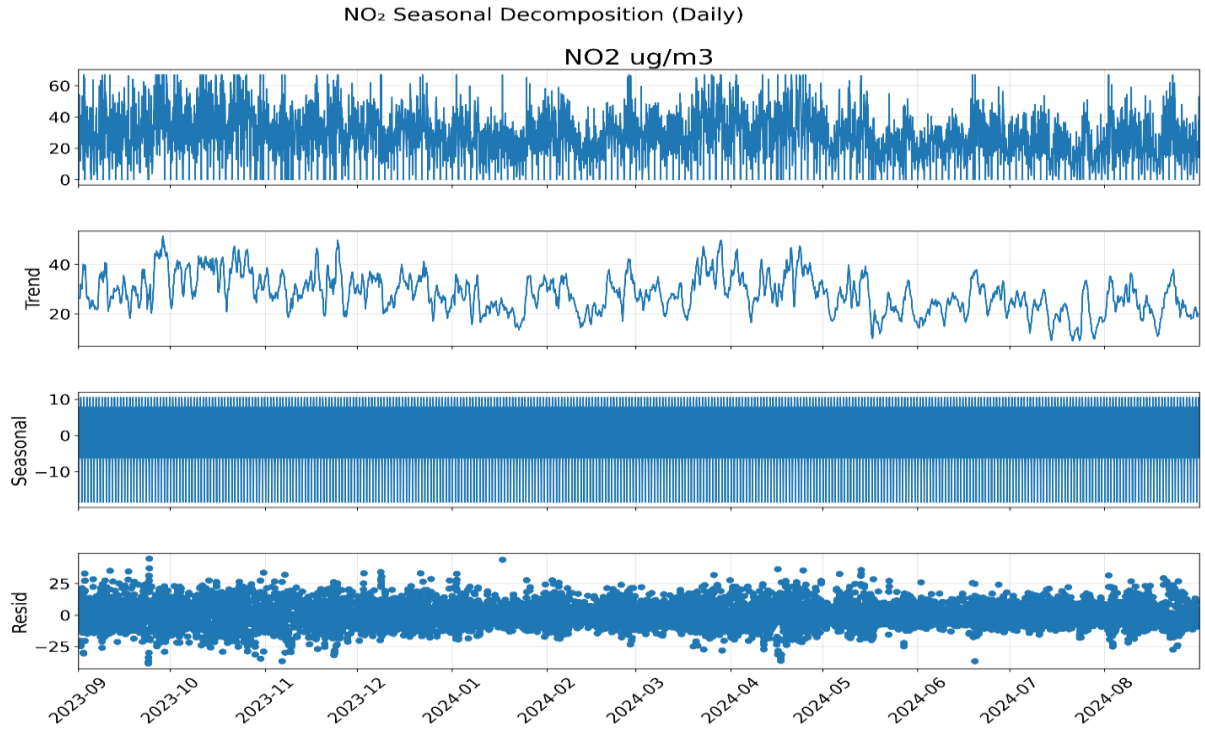


Figure C.1: Representative seasonal decomposition of the NO₂ time series (trend, seasonal, and residual components).

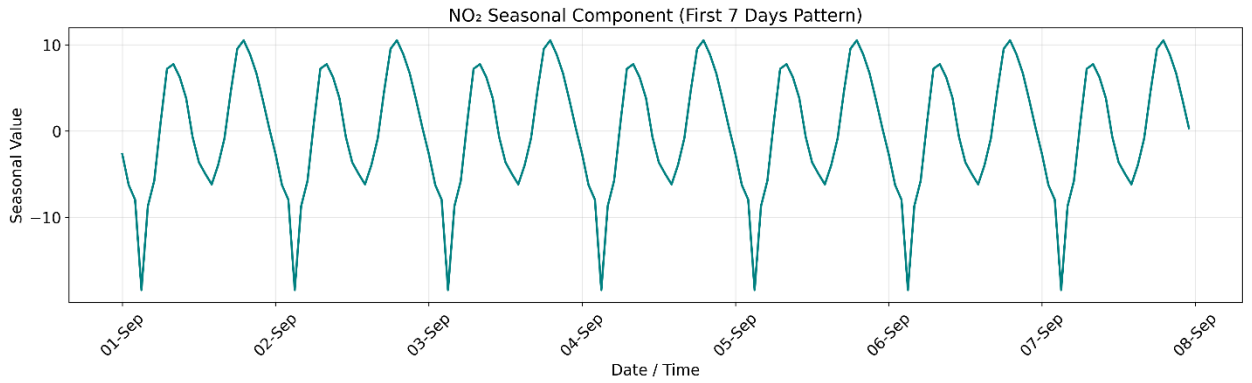


Figure C.2: Representative extracted daily seasonal pattern for NO₂ (first 7 days).

Figures C.1 and C.2 are included as representative examples of the temporal and seasonal structure identified during exploratory analysis. NO₂ was selected because it showed clear diurnal variability and is central to the forecasting comparison. Similar seasonal decomposition procedures were applied during EDA to the other pollutants, but their plots are not reproduced here to avoid unnecessary repetition.

D. Feature engineering validation

Table D.1: Numerical validation of cyclical time encoding (unit-circle constraint).

Statistic	Hour (sin/cos) Error	Day-of-Week (sin/cos) Error
Count	8784	8784
Mean	1.3878×10^{-17}	3.1838×10^{-17}
Std	4.8740×10^{-17}	6.2017×10^{-17}
Min	0.0000	0.0000
25%	0.0000	0.0000
50%	0.0000	0.0000

75%	0.0000	0.0000
Max	2.2204×10^{-16}	2.2204×10^{-16}

Table D.1 validates the sine and cosine transformations used for cyclical temporal features. The unit-circle error measures deviation from $\sin^2(\theta) + \cos^2(\theta) = 1$. The near-zero mean and maximum errors indicate that the hour-of-day and day-of-week encodings were implemented correctly and preserve cyclical continuity.